

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
UNIVERSITY OF ECONOMICS AND LAW

FACULTY OF INFORMATION SYSTEMS



ANALYSIS OF 4-STAR HOTEL QUALITY IN VUNG TAU FROM ACTUAL TOURIST EXPERIENCES THROUGH REVIEW DATA ON TRAVELOKA

Supervisor : **M.Sc. Le Ba Thien**

Group: **No.12**

Class: **252BIE402501**

HCM, March 25th, 2026

GROUP MEMBER SELF-ASSESSMENT TABLE

No.	Student ID	Full Name	Task	Completion
1	K234131503	Nguyen Thien Quoc	Web data crawling, Data Visualization	100%
2	K234131491	Nguyen Thuc Ai Ngan	Building and training Machine Learning models	100%
3	K234131494	Vuong Nguyen Hong Ngoc	Comparative analysis, Final conclusion	100%
4	K234131506	Truong Thi Trang Thi	Building and training Machine Learning models	100%
5	K234131510	Tran Thi Minh Thu	Data cleaning and processing	100%

ACKNOWLEDGEMENTS

We would like to express our sincere gratitude to our instructor, **M.Sc. Le Ba Thien**, for his dedicated guidance and support in helping us complete this final project. He assisted us in identifying the key aspects of the topic and provided insightful feedback that enabled us to refine our work and achieve the best possible results.

Furthermore, we would like to extend our deep appreciation to the researchers and authors of the studies cited in this report. Their published works have served as invaluable resources, providing our group with a deeper and more comprehensive perspective during the research process.

We sincerely thank you.

The Project Team

DECLARATION

We hereby declare that the content of this project is the result of our own serious study and research. We affirm that this work is original and does not contain any material copied from other sources without proper and specific citation.

The Project Team

TABLE OF CONTENTS

GROUP MEMBER SELF-ASSESSMENT TABLE.....	i
ACKNOWLEDGEMENTS	ii
DECLARATION	iii
TABLE OF CONTENTS	iv
LIST OF TABLES.....	vii
LIST OF FIGURES.....	viii
CHAPTER 1: INTRODUCTION	1
1.1. Rationale for the study	1
1.2. Research Objectives	2
1.3. Research Scope and Object of Study	3
CHAPTER 2: THEORETICAL FRAMEWORK AND.....	6
RESEARCH OVERVIEW	6
2.1. Overview of 4-star Hotels and Tourism in Vung Tau	6
2.2. Online Reviews and Tourist Behavior	6
2.3. Sentiment Analysis.....	8
2.3.1. Levels of Sentiment Analysis	8
2.3.2. Key Sentiment Analysis Methods	9
2.3.3. Advantages and Disadvantages of Sentiment Analysis.....	10
2.3.3.1. Advantages	10
2.3.3.2. Disadvantages.....	10
2.4. Aspects of Hotel Service Quality	11
2.5. Text Analysis Models	12
2.5.1. Overview of Support Vector Machine (SVM) and TF - IDF	12
2.5.2. Overview of PhoBERT language model	14

2.5.3. Overview of BERTopic Topic Modeling Framework and Sentence Transformer	15
CHAPTER 3: DATA AND RESEARCH METHODOLOGY	19
3.1. Data Source and Description.....	19
3.2. Data Collection Methodology	22
3.3. Data Preprocessing.....	23
3.3.1. Removing Irrelevant Variables.....	23
3.3.2. Text Translation.....	24
3.3.3. Text Normalization	25
3.3.4. Tokenization and Stopword Removal	27
3.4. Label Creation and Handling Imbalance	30
3.4.1. Label Creation (2 classes).....	30
3.4.2. Computing Class Weights.....	33
3.5. Text Feature Extraction Techniques	33
3.5.1. Data Representation using TF-IDF (for SVM).....	33
3.5.2. Data Representation using PhoBERT Tokenizer.....	36
CHAPTER 4: MODEL DEVELOPMENT AND EVALUATION	37
4.1. Sentiment Classification Model Development	37
4.1.1. Initialization and Training of the Support Vector Machine (SVM) Model..	37
4.1.2. Initialization and Fine-tuning of the PhoBERT Model	38
4.2. Evaluation and Validation Methods.....	42
4.2.1. Stratified K-Fold Cross Validation	42
4.2.2. Train - Test Split using Scikit-learn	43
4.2.3. Evaluation Metrics	44
4.3. Comparison between SVM and PhoBERT	45
4.4. Negative Aspect Extraction Method	46
4.4.1. Extracting Negative Reviews Using the Best-Performing Model.....	46

4.4.2. Applying the BERTopic Model for Topic Modeling	47
4.4.3. Semantic Similarity–Based Aspect Mapping	49
CHAPTER 5: RESULTS AND DISCUSSION	53
5.1. Sentiment Analysis Results.....	53
5.2. Identification of Bottlenecks and Service Quality Gaps.....	55
5.2.1. Service Aspects Requiring Improvement	56
5.2.2. Comparison with 5-Star Hotel Standards	59
5.3. Proposed Solutions for Service Quality Improvement	61
5.4. Limitations and Future Research Directions.....	62
5.4.1. Limitations	62
5.4.2. Future Research Directions.....	63
REFERENCES	65

LIST OF TABLES

Table 1.1: List of 4-star hotels in Vung Tau	4
--	---

LIST OF FIGURES

Figure 1.1: Illustrating the search for 4-star standard hotels on Traveloka.....	3
Figure 1.2: Customer reviews of Fusion Suites Vung Tau on Traveloka.	5
Figure 2.1: Illustration of the Elaboration Likelihood Model (ELM).....	8
Figure 3.1: The original data table used for analysis in the research project.	19
Figure 3.2: Overview Data Analysis	20
Figure 3.3: Distribution of Customer Review Scores	21
Figure 3.4: Data Collection and Extraction Method	22
Figure 3.5 : Variables After Data Cleaning.....	23
Figure 3.6: Results After Removing Duplicate Data	25
Figure 3.7: Results After Tokenization and Stopword Removal	30
Figure 3.8: Rule-Based Sentiment Classification from Review Scores.....	35
Figure 4.1: SVM Model Initialization and Training Setup	37
Figure 4.2: Results from SVM	45
Figure 4.3: Results from PhoBERT.....	45
Figure 4.4: Embedding Model Loading Warning	47
Figure 4.5: Construction of Topic DataFrame with Keywords and Mapped Aspects ..	51
Figure 5.1: Word Cloud Visualization of Key Aspects Across Service Categories	54
Figure 5.2: Frequency distribution of tourists' negative reviews by aspect	56

CHAPTER 1: INTRODUCTION

1.1. Rationale for the study

Following the Covid-19 pandemic, the Vietnamese tourism industry has witnessed an impressive recovery with breakthrough growth. According to data from the Vietnam National Authority of Tourism, the arrival of over 108 million domestic tourists in 2023 exceeded the set targets and contributed significantly to the total revenue of the service and tourism sector, estimated at over 670 trillion VND. This result confirms the role of tourism as a key economic sector and demonstrates a clear increase in the scale and demand for domestic tourism consumption. The rapid increase in visitor numbers not only provides momentum for the development of various travel activities but also drives the need for expanding and improving the quality of support services, particularly accommodation, to better meet the increasingly diverse experiences and expectations of travelers. In this context, Vung Tau city, with its strategic geographical advantage and developed tourism infrastructure, has become a top priority destination, especially for the market of tourists from Ho Chi Minh City and neighboring provinces.

In Vung Tau, the 4-star hotel segment accounts for a significant proportion of the accommodation structure and is a popular choice for many tourists due to the balance between cost and service quality. With reasonable pricing while still ensuring standards for amenities and services, this segment effectively meets the needs of mid-range and upper-mid-range tourist groups. Data from the online booking platform Traveloka shows that many 4-star hotels in Vung Tau have received tens of thousands of reviews from guests, with average scores ranging from approximately 8.0 to 8.8 on a 10-point scale. While the service quality of hotels in this segment is generally considered quite good, these ratings remain relatively low compared to 5-star hotels (which typically achieve average scores of 9.0/10 or higher on online review platforms), thereby reflecting a certain disparity in service standards and customer experience between the two segments.

Relying solely on hardware rating standards, such as room area or the number of amenities, is insufficient to explain why many 4-star hotels fail to satisfy demanding customer segments. Therefore, conducting research based on actual user review data

from the Traveloka travel platform will provide the most objective perspective. Instead of general assessments, applying Machine Learning models for Sentiment Analysis and Aspect Extraction will help accurately identify the 'bottlenecks' that are dragging down the scores of these hotels.

On this basis, the team has selected the topic “Analysis of 4-star hotel quality in Vung Tau from actual tourist experiences through review data on Traveloka” with the desire to deeply analyze actual tourist reviews to find the root causes of declining guest experiences. To identify the true factors reducing tourist satisfaction, the study focuses on an in-depth analysis of negative reviews. By comparing these limitations with the rigorous service standards of the 5-star segment, the research expects to explain the core reasons hindering the advancement of 4-star hotels in Vung Tau. The research results will serve as an important basis for hotel managers to re-evaluate their service processes, thereby proposing solutions to improve and enhance quality in line with tourist needs. This will not only help improve the competitiveness of accommodation establishments but also contribute to elevating the Vung Tau tourism brand in the context of increasingly stringent tourist expectations.

1.2. Research Objectives

In the smoke-free industry, the success of a lodging establishment is not solely measured by pure financial indicators but also lies in the ability to balance between maximizing profit and satisfying the expectations of visiting customers. Instead of seeking a one-size-fits-all formula, modern hotel businesses are gradually shifting towards consolidating values from practical experiences. Continuously listening and enhancing customer experiences based on proven management methods is the true “key” that helps hotels maintain their competitive edge. Stemming from this trend, this study is conducted by deeply researching data from Traveloka, with the main objective of evaluating the service quality status of 4-star hotels in Vung Tau based on data extracted from actual traveler experiences on the Traveloka platform. Subsequently, an automatic sentiment classification model will be proposed, along with management methods to enhance competitive capabilities for the local lodging industry.

To achieve the overall objective, the research focuses on addressing the following five issues:

- (1) Data Collection and Processing:** Collecting reviews, ratings, and related information of 4-star hotels in Vung Tau on Traveloka to form an analytical database.
- (2) Analysis of Psychological and Behavioral Characteristics of Travelers:** Identifying the factors that travelers are most concerned about through the frequency of keywords in reviews.
- (3) Construction and Testing of Machine Learning Models:** Applying Support Vector Machine (SVM) and PhoBERT algorithms to classify traveler sentiments (Positive, Negative, Neutral) with optimal accuracy.
- (4) Visualization of Service Quality:** Using dashboard to represent the dynamics of service quality over time, across different customer segments, and in various specific service aspects.
- (5) Proposal of Solutions:** Based on the analysis results, providing recommendations to help hotel managers address weaknesses and leverage strengths to maximize customer experience.

1.3. Research Scope and Object of Study

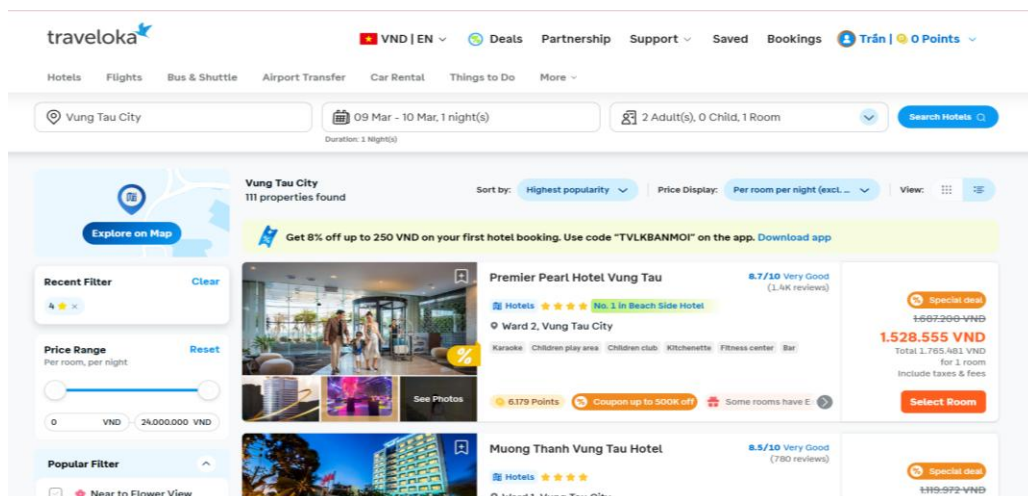


Figure 1.1: Illustrating the search for 4-star standard hotels on Traveloka.

The research scope is focused on 4-star standard hotels in Vung Tau city with publicly available reviews on the Traveloka online booking platform. The data for the

study is collected directly from this platform, including reviews in both Vietnamese and other languages posted by customers in recent years on the platform's comment section. This ensures the research is based on authentic and current feedback from guests, allowing for a comprehensive analysis of customer experiences and satisfaction levels at these establishments. If you have any more details or questions regarding the research scope, feel free to ask.

Table 1.1: List of 4-star hotels in Vung Tau

No.	Hotel
1	Muong Thanh Vung Tau Hotel
2	Premier Pearl Hotel Vung Tau
3	The Cap Hotel
4	Petro Hotel
5	DIC Star Landmark VT Hotel & Apartment
6	Sammy Hotel VT
7	petro House VT
8	Fusion Suites VT
9	Minera Hot Springs Binh Chau
10	Grand Hotel VT
11	Palace Hotel VT
12	Iconic Long Hai Hotel
13	Holiday Inn Resort HO TRAM BEACH by IHG

The research subject of the study is the content of customer reviews, including feedback on room quality, staff service attitude, amenities, location, cleanliness, and

other services. Through analyzing the content and sentiments in these comments, the research aims to evaluate customer satisfaction levels and identify both positive aspects and limitations in the service quality of the 4-star hotels in Vung Tau.

Đánh giá của khách hàng

**Khách đã xác minh**
 Kỳ nghỉ tại chỗ

10,0 /10

Đánh giá cách đây 8 tuần

Khách sạn có nhiều view đẹp,decor rất tỉ mỉ các góc,đi vài bước chân là ra đến bãi trước



 [Đánh giá này hữu ích không?](#)

**Khách đã xác minh**
 Kỳ nghỉ tại chỗ

9,1 /10

Đánh giá cách đây 9 tuần

Không gian tuyệt vời, thực đơn buffet phong phú.

translated by Google [Xem bản gốc](#)

 [Đánh giá này hữu ích không?](#)

**VD** **Vuong D.**
 Giải trí

9,4 /10

Đánh giá cách đây 18 tuần

Bữa sáng ngon, đa dạng, thích hợp cho trẻ con lẫn người lớn

 [Đánh giá này hữu ích không?](#)

Figure 1.2: Customer reviews of Fusion Suites Vung Tau on Traveloka.

CHAPTER 2: THEORETICAL FRAMEWORK AND RESEARCH OVERVIEW

2.1. Overview of 4-star Hotels and Tourism in Vung Tau

Amid the wave of globalization and the Fourth Industrial Revolution, digital technology has assumed a pivotal, central role, creating the foundation and momentum for remarkable advances in the socio-economic landscape. This wave requires all sectors of society to become digitized, and the tourism and hospitality industry is no exception. Online booking platforms have surged, giving rise to a new concept in which hotel quality is no longer confined to administrative classification standards such as the Vietnamese National Standard TCVN 4391:2015 on Hotel Rating (2015). Instead, a new trend has emerged: perceived quality, where the actual star rating of an accommodation is defined by the experiences and satisfaction of guests themselves.

On the Traveloka platform, the segment of hotels equivalent to 4-star standards is not merely composed of establishments with high-end facilities, but rather those that achieve a rating threshold of 8.0/10 or higher based on thousands of real customer reviews, reflecting a direct correlation between the cost incurred and the value received.

In the Vung Tau market, which receives hundreds of thousands of visitors during peak holidays such as April 30th to May 1st, 2025, when Ba Ria - Vung Tau province welcomed 858,000 visitors - the level of competition is extremely high, and the flow of tourists fluctuates seasonally. Therefore, a hotel's ability to maintain a "4-star" rating on digital platforms serves as evidence of strong operational capacity and an exceptional ability to meet customer expectations. Focusing analysis on this group of hotels, rather than relying solely on administrative licensing, enables a deeper understanding of market preferences, thereby transforming qualitative sentiments into meaningful variables within an economic model.

2.2. Online Reviews and Tourist Behavior

Online reviews are comments, ratings, and feedback posted by customers on online platforms after experiencing products or services. In the tourism industry, these reviews reflect the actual satisfaction levels of tourists with hotels through ratings,

review content, and lodging experiences. For the 4-star hotel segment, online reviews become a crucial source of information for tourists to reference and compare service quality before making booking decisions.

Tourist behavior in the online environment is often demonstrated through information searching, evaluating options, and decision-making based on collected internet information. Before selecting a hotel, tourists typically consult online reviews to gain a better understanding of room quality, services, amenities, and the real experiences of previous customers. This information helps tourists minimize uncertainty and risks in choosing lodging services, shaping perceptions and attitudes toward hotels. Therefore, online reviews can significantly influence the decision-making process and booking intentions of tourists.

According to the Elaboration Likelihood Model, online reviews influence tourist behavior through two information reception routes. The central route involves detailed review content such as room quality, services, and amenities, while the peripheral route includes factors like average ratings, review quantity, and platform credibility.

Additionally, the Uses and Gratifications theory suggests that users actively choose media or online platforms to satisfy specific personal needs. In the context of online tourism, tourists use booking platforms to search for information, compare options, and evaluate service quality before making decisions. By reading and referencing online reviews, tourists can reduce risks in service selection, increase confidence in booking decisions, and enhance satisfaction with their choices. Thus, online reviews not only reflect post-consumption experiences but also play a crucial role in guiding tourists' hotel selection behaviors.

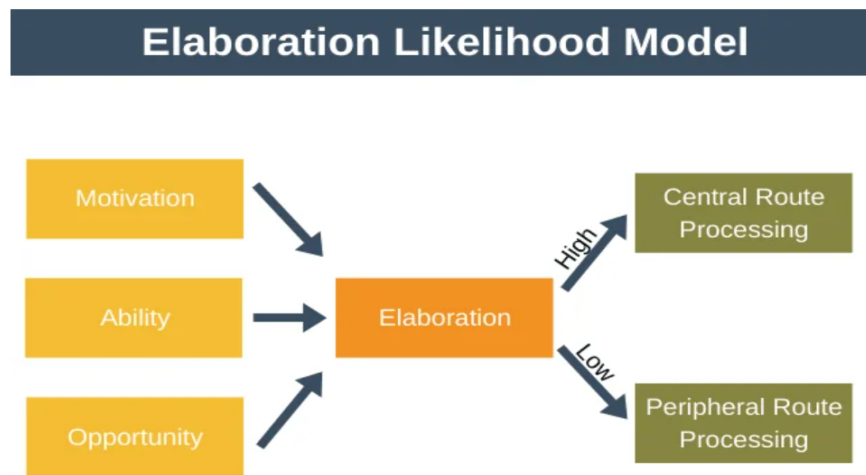


Figure 2.1: Illustration of the Elaboration Likelihood Model (ELM)

However, in practice, tourists' reviewing behavior does not occur uniformly. Most users tend to leave reviews only when their experiences are extremely satisfying or highly dissatisfying. Experiences that are moderate or average are less frequently shared because they do not generate sufficient motivation for users to spend time writing a review. This phenomenon leads to a skewed distribution of ratings, where positive and negative reviews account for a larger proportion than neutral ones. Neutral cases are relatively rare and often do not convey clear emotional polarity. Therefore, focusing on positive and negative reviews while excluding neutral cases becomes a reasonable approach for the classification task. This approach not only aligns with the empirical characteristics of online review datasets but also allows the model to better concentrate on distinguishing between two clearly defined sentiment states.

2.3. Sentiment Analysis

Sentiment Analysis is an important branch in the field of Natural Language Processing (NLP) that aims to identify and classify the emotions, attitudes of the writer towards an object such as a product, service, etc. Sentiment Analysis plays a crucial role in various fields, including tourism, shopping, accommodation services, where customer online feedback becomes a valuable data source for evaluating service quality.

2.3.1. Levels of Sentiment Analysis

Sentiment Analysis consists of three main levels: Sentence-level, Document-level, and Aspect-level. At the sentence level, it helps identify the emotions expressed in individual sentences. Each sentence is treated as an independent analysis unit and classified as positive, negative, or neutral. However, at the sentence level, it may not

accurately determine the emotions associated with specific aspects of the evaluated object. On the other hand, document-level sentiment analysis focuses on determining the overall sentiment of the entire document based on the overall content. The limitation of this method is that it does not reflect the emotional differences between different aspects within the same document. Meanwhile, at the aspect level, which is the most detailed analysis level, it not only identifies emotions but also clarifies which aspect the emotion is directed towards.

In the scope of this research, sentiment analysis is implemented at the aspect level to determine customer emotions towards specific factors influencing the service quality of hotels. This helps provide a detailed insight into the strengths and limitations of each factor.

2.3.2. Key Sentiment Analysis Methods

Sentiment Analysis employs various approaches, which can be divided into three main groups: Rule-based methods, Machine learning-based methods, and Deep learning-based methods.

Rule-based sentiment analysis uses a set of grammatical rules and sentiment dictionaries to analyze text. The system scans each word, phrase, and compares it with a labeled sentiment dictionary (positive, negative, and neutral). For example, in the sentence "The scenery here is beautiful," the system will identify the word "beautiful" as positive and label the sentiment of the sentence as positive. Conversely, if the sentence contains words like "ugly" or "bad," it will be classified as negative. However, the limitation of this method is the difficulty in handling complex contextual cases such as negations, sarcasm, or ambiguity.

On the other hand, machine learning methods utilize labeled sentiment data for model training and classification. After learning from this data, the model can automatically classify sentiments in new texts more accurately. Some common algorithms include Naive Bayes, Logistic Regression, etc. Compared to rule-based methods, traditional machine learning methods often achieve higher accuracy. However, the model's effectiveness depends significantly on the quality of training data and feature extraction process.

Deep learning methods use neural network models to automatically learn features from text data without requiring many manual processing steps. Popular models like Recurrent Neural Network, Long Short-Term Memory, have better contextual processing capabilities, especially with long and complex texts. However, deep learning methods require significant computational resources and large datasets to achieve optimal effectiveness.

2.3.3. Advantages and Disadvantages of Sentiment Analysis

2.3.3.1. Advantages

First and foremost, sentiment analysis helps aggregate a large amount of data from various sources such as social media, product reviews, customer emails, and service feedback. As a result, businesses have a more comprehensive view of customer attitudes towards the brand. Additionally, sentiment analysis supports real-time monitoring of customer feedback. With the ability for quick and automated processing, businesses can promptly detect negative feedback, issues in service delivery, and take appropriate corrective measures to enhance customer experience and increase brand loyalty.

Furthermore, applying sentiment analysis contributes to automating customer care processes. Artificial intelligence systems can classify feedback by sentiment and automatically direct customers to frequently asked questions or appropriate support departments. Lastly, sentiment analysis helps improve marketing strategies by analyzing customer responses to advertising campaigns. Businesses can adjust marketing messages to optimize effectiveness. Understanding customer psychology helps build more relevant content and communication strategies.

2.3.3.2. Disadvantages

Firstly, systems face challenges in accurately determining neutral sentiments or comments with unclear content. Descriptive statements, such as product feature comments, may be classified as neutral, while users are indirectly expressing dissatisfaction. This increases the risk of bias in sentiment classification results. Additionally, sentiment analysis models often struggle to identify context and factors like sarcasm or irony. For example, a sentence like “Excellent view, the system crashed

just when I needed to pay.” might be classified as positive due to the word “excellent,” affecting the accuracy of the analysis results.

Another limitation is the difficulty in processing data with complex language. Factors such as abbreviations, spelling errors, emoticons, and evolving language usage on social media make it challenging for algorithms to keep up. Lastly, despite advancements in artificial intelligence technology, sentiment analysis systems still require human intervention and monitoring to ensure accuracy. This reduces the level of automation and increases operating costs.

2.4. Aspects of Hotel Service Quality

In the context of online review analysis (E-WOM), particularly on the Traveloka platform, tourists tend to directly reflect their actual experiences and perceptions at the time of their stay, rather than comparing the gap between expectations and perceptions as in the SERVQUAL model. Therefore, this study adopts the improved model proposed by Cronin & Taylor (1992), namely SERVPERF, which measures service quality based on actual performance and customers’ perceived experiences.

However, to operationalize the SERVPERF dimensions into practical criteria within the context of accommodation services in Vietnam, this study aligns these components with the Vietnamese National Standard TCVN 4391:2015 on Hotel Classification. Within the SERVPERF model, tangibility represents physical facilities, equipment, and staff appearance. When aligned with TCVN 4391:2015, this group of factors is strictly regulated and divided into specific aspects that tourists directly interact with:

- **Facilities:** Corresponding to standards on “Location and Architecture”, including spatial design, landscape, lobby areas, parking, and internal transportation systems within the hotel.
- **Amenities:** Referring to standards related to interior quality, guest rooms, air conditioning systems, sanitary equipment, Wi-Fi, and additional recreational services (e.g., swimming pools, spas).
- **Cleanliness:** TCVN 4391:2015 establishes an independent and rigorous set of hygiene standards, requiring absolute cleanliness in public areas, guest rooms,

and air quality. This is the most critical tangible factor influencing negative online customer sentiment.

- **Food:** Corresponding to standards on restaurant and bar quality, as well as food safety and hygiene, contributing to the overall physical service experience.

The remaining SERVPERF dimensions-reliability, responsiveness, assurance, and empathy-primarily reflect human and process-related aspects, including service performance, willingness to assist, professional knowledge, and personalized care. When mapped to TCVN 4391:2015, these psychological and behavioral factors are directly represented through the standards for “Management and Service Staff” and “Service Quality and Level of Service”. Therefore, these dimensions are consolidated into a single variable: Staff/Service, which evaluates actual performance through staff professionalism, foreign language communication skills, service attitude, responsiveness, and problem-solving capabilities during customer interactions.

The integration of performance-based measurement (SERVPERF) with industry-standard evaluation frameworks creates a comprehensive analytical model. This serves as the theoretical foundation for guiding, validating, and interpreting the latent topics extracted from the BERTopic algorithm. In subsequent analysis, the identification and grouping of factors contributing to negative reviews will be closely aligned with this framework, ensuring that the model’s results are not only algorithmically robust but also practically relevant to hotel service quality management.

2.5. Text Analysis Models

2.5.1. Overview of Support Vector Machine (SVM) and TF - IDF

**** Support Vector Machine Model***

Support Vector Machine is a powerful classification method developed based on the idea of maximizing the margin between classes. Unlike Logistic Regression, SVM seeks a hyperplane such that the distance from the hyperplane to the nearest data points of each class is maximized. SVM was originally developed by Vapnik (1963) primarily for binary classification problems and was later extended to multi-class classification. One of the outstanding characteristics of SVM is its ability to handle high-dimensional data, making it well-suited for text analysis.

Consider a binary classification problem with a training dataset (x_i, y_i) , where x_i is the feature vector and $y_i \in \{-1, +1\}$ is the class label. SVM finds a hyperplane of the form:

$$w^T * x + b = 0$$

to divide the feature space into two regions corresponding to the two classes. The distance from a data point to the hyperplane is determined along the perpendicular direction, and the margin of the model is the minimum distance between the data points of each class and the hyperplane. The objective of SVM is to find parameters w and b such that the margins of the two classes are equal and reach the maximum value. The above problem is transformed into a convex optimization problem with the objective function being the minimization of the squared norm of the weight vector $\|w\|^2$, subject to the classification constraint $y_i(w^T * x + b) \geq 1$. This optimization ensures that the obtained solution is the unique global optimum, helping SVM avoid the issue of getting stuck in local minima, which is common in nonlinear learning models.

In text analysis, data is often not completely linearly separable. Therefore, SVM is extended through the concept of a soft margin, which allows some data points to violate the classification conditions in order to achieve a reasonable trade-off between accuracy on the training set and generalization ability. The degree of tolerance for misclassification is controlled through the parameter C . When the value of C is larger, the model prioritizes fitting the training data but is more prone to overfitting; conversely, when C is smaller, it creates a wider margin and the model becomes more stable on new data. Typically, documents and texts are represented by feature vectors with very high dimensionality, sometimes even larger than the number of observations. SVM still performs effectively in this case because the complexity of the model depends on the number of support vectors rather than directly on the dimensionality of the data. As a result, SVM has become one of the most commonly used models and achieves high performance in traditional text classification tasks.

**** Term Frequency - Inverse Document Frequency (TF - IDF) Technique***

The TF - IDF technique is developed based on the IDF method proposed by Sparck Jones (1972, 2004) with a fundamental principle: if a query keyword appears in too many different documents, then that keyword has lost its distinguishing value in

helping differentiate content among documents. For this reason, the model “penalizes” overly common words (such as conjunctions and filler words) by assigning them very low weights. In contrast, rare words or domain-specific terms that appear in only a few documents are assigned higher weights because they provide the most valuable information for identifying the content of that document.

This idea is consistent with the language modeling theory of Robertson (2004), which suggests that terms in a given document can be divided into two types: elite terms and non-elite terms, thereby measuring the importance of a term within a collection of documents.

The TF - IDF formula is calculated by multiplying two values:

$$w_{i,j} = tf_{i,j} \times \log \left(\frac{N}{df_i} \right)$$

Where:

- + $w_{i,j}$: The TF-IDF score (weight) of term i in document j .
- + N : The total number of documents in the entire dataset.
- + $tf_{i,j}$: The frequency of term i in document j .
- + df_i : The number of documents that contain term i .

2.5.2. Overview of PhoBERT language model

PhoBERT (“Pho” refers to phở, a famous Vietnamese food) is the first large-scale language model specifically designed for the Vietnamese language. It is developed based on the BERT architecture and further optimized following the training approach of RoBERTa. Unlike multilingual models, PhoBERT is specifically tailored for Vietnamese, allowing it to achieve superior performance compared to the strongest multilingual model, XLM-R.

PhoBERT is released in two versions, namely *PhoBERT_{base}* and *PhoBERT_{large}*, which share the same architectural configurations as *BERT_{base}* and *BERT_{large}*. However, they are optimized using RoBERTa’s training strategies to improve model performance and stability. Specifically, PhoBERT removes the Next Sentence Prediction (NSP)

objective and focuses solely on the Masked Language Modeling (MLM) task, while also being trained on a larger and more diverse corpus.

The model is trained on a Vietnamese text corpus of approximately 20GB, including 1GB from Vietnamese Wikipedia and 19GB from news sources. This large-scale dataset enables PhoBERT to capture a wider range of real-world linguistic contexts compared to models trained solely on Wikipedia data.

A key distinction between PhoBERT and multilingual models lies in its word-level representation rather than syllable-level processing. To achieve this, the authors employ the RDRSegmenter tool for word segmentation prior to training, followed by the application of the FastBPE algorithm to construct subword units with a vocabulary size of approximately 64,000 tokens. This approach allows the model to effectively handle rare and out-of-vocabulary words.

From a technical perspective, PhoBERT is built upon the Transformer architecture with a self-attention mechanism, enabling it to learn bidirectional contextual relationships between words in a sentence. After pre-training, PhoBERT can be easily fine-tuned for various natural language processing tasks, such as text classification, sentiment analysis, named entity recognition, and natural language inference, by adding a classification layer on top of the model and training it on labeled data.

2.5.3. Overview of BERTopic Topic Modeling Framework and Sentence Transformer

**** BERTopic Topic Modeling Framework***

BERTopic is a modern topic modeling approach that combines embedding-based text representations with Transformer architectures to identify latent topics within a corpus. Unlike traditional models such as Latent Dirichlet Allocation (LDA), which rely primarily on probabilistic assumptions and word frequency distributions, BERTopic focuses on capturing the semantic meaning of text. By leveraging semantic embeddings, the model can identify documents with similar meanings even when different vocabularies are used.

The BERTopic framework consists of four main stages:

- BERT Embedding: This stage utilizes Transformer-based models such as BERT, RoBERTa or PhoBERT to convert textual data into dense numerical vectors (embedding vectors). Each document is represented as a high-dimensional vector in semantic space, capturing its contextual meaning. Unlike traditional approaches such as Bag-of-Words, which rely solely on word frequency, BERT produces contextualized embeddings, meaning that the representation of a word varies depending on its surrounding context. Furthermore, BERT is a bidirectional Transformer model, allowing it to capture context from both preceding and succeeding tokens in a sentence. As a result, the generated embeddings provide a deeper and more accurate semantic representation compared to earlier methods.

- Dimensionality Reduction (UMAP): Since the embedding vectors generated in the previous step are typically high-dimensional (e.g. 768 dimensions for BERT), dimensionality reduction is necessary to improve computational efficiency and enhance clustering performance. BERTopic employs UMAP (Uniform Manifold Approximation and Projection), a nonlinear dimensionality reduction technique. UMAP constructs a nearest-neighbor graph in the original high-dimensional space and then projects it into a lower-dimensional space while preserving the local and global structure of the data. According to McInnes et al. (2018), UMAP not only maintains the intrinsic structure of the data more effectively than some traditional methods but also offers high computational efficiency, making it suitable for large-scale and high-dimensional datasets.

- Clustering (HDBSCAN): After dimensionality reduction, BERTopic applies clustering to group similar documents and identify latent topics. The model uses HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise), a density-based clustering algorithm. HDBSCAN groups data points based on their density, forming clusters in high-density regions while labeling low-density points as noise (outliers). Unlike many clustering algorithms, HDBSCAN does not require a predefined number of clusters, making it highly flexible for real-world data. Additionally, it can detect clusters with varying densities by constructing a hierarchical structure and selecting the most stable clusters.

- Topic Extraction (c-TF-IDF): The final stage involves extracting representative topics for each cluster. BERTopic employs class-based TF-IDF (c-TF-IDF), a modified version of the traditional TF-IDF method tailored for topic modeling. Instead of computing TF-IDF on individual documents, c-TF-IDF treats each cluster as a single aggregated document by concatenating all texts within the same cluster. It then calculates the importance of each term based on two factors: its frequency within the cluster and its distinctiveness across all clusters. Specifically, words that appear frequently in one topic but rarely in others receive higher c-TF-IDF scores and are selected as representative keywords for that topic.

$$c - TF - IDF_{t,c} = TF_{t,c} \times \log\left(1 + \frac{A}{f_t}\right)$$

$TF_{t,c}$: Frequency of word t in class c

A : Average number of words per class.

f_t : Frequency of word t across all classes.

BERTopic is a modern topic modeling approach that effectively integrates semantic representation through Transformer-based models, dimensionality reduction techniques, and density-based clustering algorithms. This combination not only enables the discovery of latent topics within a text corpus but also ensures that the extracted topics are highly interpretable and semantically coherent with the underlying data. Compared to traditional methods such as Latent Dirichlet Allocation (LDA), BERTopic demonstrates superior performance in capturing contextual information and handling complex natural language data. As a result, the model has been increasingly adopted in various text analysis applications, including customer review analysis, social media mining, and market research.

**** Sentence Transformer***

Sentence Transformers are a method for generating sentence-level and document-level embeddings based on the Transformer architecture, designed to improve embedding quality at the sentence level compared to models such as BERT. In traditional BERT models, embeddings are primarily optimized at the token level and are not explicitly designed for measuring semantic similarity between sentences. As a result,

using BERT directly for sentence similarity tasks is often inefficient and computationally expensive.

To address this limitation, Sentence Transformers employ Siamese or Triplet Network architectures, where multiple sentences are passed through the same Transformer model to produce embeddings within a shared vector space. The training objective is to ensure that semantically similar sentences are mapped to vectors that are close to each other in the embedding space, while dissimilar sentences are positioned farther apart. Specifically, each input sentence is encoded into a fixed-length vector representation, which can then be used to compute semantic similarity between sentences using cosine similarity. This mechanism enables faster and more accurate sentence comparisons compared to standard BERT approaches.

According to the study by Nils Reimers and Iryna Gurevych (2019), Sentence Transformers not only significantly improve computational efficiency but also enhance embedding quality in tasks such as semantic similarity, clustering, and information retrieval. Consequently, this approach has been widely adopted in natural language processing applications, particularly in systems such as BERTopic, where it is used to generate embeddings prior to topic modeling.

CHAPTER 3: DATA AND RESEARCH METHODOLOGY

3.1. Data Source and Description

STT	reviewId	productType	reviewer	reviewTimestamp	ReviewScore	reviewMaxScore	reviewContentText	reviewOriginalText	translationStatus	
0	1	1856073066124162992	HOTEL	{'reviewId': '328133006', 'reviewerName': 'D...'	1.770089e+12	10.0	10.0	nhân viên rất nhiệt tình, phòng sạch sẽ, buffe...	nhân viên rất nhiệt tình, phòng sạch sẽ, buffe...	ORIGINAL
1	2	1855265970105368446	HOTEL	{'reviewId': '370266492', 'reviewerName': 'T...'	1.769320e+12	9.4	10.0	Chăm sóc khách hàng thân thiện và chuyên nghiệp	Chăm sóc khách hàng thân thiện và chuyên nghiệp	ORIGINAL
2	3	1855075484958738747	HOTEL	{'reviewId': '396809345', 'reviewerName': 'N...'	1.769138e+12	10.0	10.0	Nhân viên thân thiện và nhiệt tình	Nhân viên thân thiện và nhiệt tình	ORIGINAL
3	4	1854849274851903803	HOTEL	{'reviewId': '335212160', 'reviewerName': 'V...'	1.768922e+12	10.0	10.0	Phòng đẹp và view đẹp, giường êm mát, tiện ích...	Phòng đẹp và view đẹp, giường êm mát, tiện ích...	ORIGINAL
4	5	1854529681735302882	HOTEL	{'reviewId': None, 'reviewerName': 'D***h', ...}	1.768617e+12	10.0	10.0	Một địa điểm tuyệt vời để đi du lịch. Mọi thứ ...	Nice place for a trip. Everything is good. All...	GOOGLE_TRANSLATE

Figure 3.1: The original data table used for analysis in the research project.

The dataset used in this study includes customer reviews of 4-star segment hotels in Vung Tau published on the Traveloka online booking platform. These reviews reflect customers' real experiences after using accommodation services and are presented in the form of scores along with comment content.

The original dataset (untreated) consists of 2.103 records, with each record corresponding to a customer review. The initial data is stored in a table format with the following variables: STT, reviewId, productType, reviewer, reviewTimestamp, ReviewScore, reviewMaxScore, reviewContentText, reviewOriginalText, translationStatus, and some additional variables related to customer reviews during the data scraping process.

Specifically:

- reviewId is the unique identifier for each review.
- productType indicates the type of service (HOTEL).
- reviewer contains reviewer identification information (personal information is anonymized).
- reviewTimestamp represents the review timestamp in date format.
- ReviewScore is the customer's score for the hotel on a 10-point scale.
- reviewMaxScore is the maximum score on the scale (10 points).
- reviewContentText is the displayed comment content.

- reviewOriginalText is the original content of the comment (in case of translation).
- translationStatus indicates the translation status (e.g., ORIGINAL or GOOGLE_TRANSLATE).

In this research scope, the variables directly used from the original data for analysis include ReviewScore and reviewContentText. Specifically, ReviewScore is an important quantitative variable to measure satisfaction levels and conduct descriptive statistics on score distribution. Meanwhile, reviewContentText is a qualitative variable reflecting detailed customer perceptions on factors such as staff attitude, room quality, amenities, location, meals, etc. The remaining variables mainly serve to identify records and provide contextual information (time, service type, translation status).

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2103 entries, 0 to 2102
Data columns (total 20 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   STT                                   2103 non-null   int64
 1   reviewId                             2103 non-null   object
 2   productType                           2103 non-null   object
 3   reviewer                             2103 non-null   object
 4   reviewTimestamp                       2103 non-null   float64
 5   ReviewScore                           2103 non-null   float64
 6   reviewMaxScore                        2103 non-null   float64
 7   reviewContentText                     2103 non-null   object
 8   reviewOriginalText                    2101 non-null   object
 9   translationStatus                     2103 non-null   object
10   translationProviderLabelUrl           130 non-null    object
11   reviewMedia                           2103 non-null   object
12   reviewReply                           975 non-null    object
13   reviewAdditionalInformation            679 non-null    object
14   subRatingInformations                 2103 non-null   object
15   reactionSummary                       2103 non-null   object
16   reviewOrigin                          2103 non-null   object
17   reviewProvider                        2103 non-null   object
18   reviewOriginLabel                     326 non-null    object
19   reviewIconUrl                         1777 non-null   object
dtypes: float64(3), int64(1), object(16)
memory usage: 328.7+ KB
```

Figure 3.2: Overview Data Analysis

After conducting an overview data check, the results indicate that the dataset consists of 2,103 observations with a total of 20 attributes. There are no missing values in important variables such as ReviewScore and reviewContentText.

In terms of data types, the dataset includes 3 floating-point variables (float64), 1 integer variable (int64), and 16 object variables, with most of them being text data. This indicates that the dataset is suitable for natural language processing (NLP) tasks and sentiment analysis. Some variables like reviewReply, reviewAdditional Information, or

translationProviderLabelUrl have missing values; however, as these variables are not used in the model building process, they do not impact the research results.



Figure 3.3: Distribution of Customer Review Scores

The distribution chart of customer review scores reveals that the majority of customers rate between 8 and 10, with the highest proportions at ratings of 9 and 10. The average score of 8.67 indicates a relatively high level of overall satisfaction with the 4-star hotels in Vung Tau. The median score also falls at a high level, reflecting positive customer evaluations of service quality.

However, there still exists a small group of reviews with low ratings (below 6). Although the proportion is small, these reviews are essential as they highlight underlying issues in service delivery. The data distribution tends to skew left, meaning higher scores are predominant, with a long tail extending towards lower values.

The results suggest that the hotel system is operating efficiently and receiving positive feedback. However, a deeper analysis of negative reviews using machine learning models is necessary to identify specific causes and propose data-driven solutions to enhance service quality.

3.2. Data Collection Methodology

```
import subprocess

curl_cmd = r'''
curl https://www.traveloka.com/api/v2/ugc/review/consumption/v2/getReviews \
-H 'accept: */*' \
-H 'accept-language: vi-VN,vi;q=0.9,fr-FR;q=0.8,fr;q=0.7,en-US;q=0.6,en;q=0.5' \
-H 'content-type: application/json' \
-b '_fbp=0.1.176961419266.4480418024898248; _gcl_rw=1.1.593484946.1769614408; _tt_enable_cookie=1; _ttp=8UXGK79PQ918Y1P8DIAV727_tt.1; _fid=3p662stAF770y0000Y.1769614438725; _cs_ex=1768665884; _cs_c=1; __t_cld=f64ccbea-78a1-4662-8b67-1888676c4542; _jsu_yjs=1769614421.6836347-c877-4182-8af \
-H 'origin: https://www.traveloka.com' \
-H 'priority: u=1, l' \
-H 'referer: https://www.traveloka.com/vi-vn/hotel/detail?spec=45-42-2826-86-42-2826.1.1.HOTEL-9000000961953.Premier120Pear1120Hotel1120Vang1207au.20LogisPromo-10preSearch10-185029994838749922&context=07822inventoryRateKey/12733/122pov646312200cpd1289480803287top8Phe618z7277070e3121c38949546Fu \
-H 'sec-ch-ua: "Not(A;Brand)"v="8", "Chromium"v="144", "Google Chrome"v="144"' \
-H 'sec-ch-ua-mobile: ?0' \
-H 'sec-ch-ua-platform: "Windows"' \
-H 'sec-fetch-dest: empty' \
-H 'sec-fetch-mode: cors' \
-H 'sec-fetch-site: same-origin' \
-H 'te-tr: 26222' \
-H 'tv-client-sessionid: T1-web.850qvcYCM459P908RT2KCS2' \
-H 'tv-country: VN' \
-H 'tv-currency: VND' \
-H 'tv-language: vi_VN' \
-H 'tv-mcc-id: 850qvcYCM459P908RT2KCS2' \
-H 'user-agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.8.0 Safari/537.36' \
-H 'www-app-version: release_webad_28208129-2829549eb' \
-H 'x-client-interface: desktop' \
-H 'x-did: H0F12L10H0U0U0FASRTH01K0D0U0V' \
-H 'x-domain: ugc/review' \
-H 'x-route-prefix: vi-vn' \
--data-raw '{"fields":[{"objectid":"9000000961953","producttype":"HOTEL","configid":"REV_CONSIG_HOTEL_GENERAL_VN","ratingtarget":1},"origin":"TRAVELOKA","skip":"W","limit":"40","sort":"SORT_CREATED_DESCENDING","filter":{"format":"FORMAT_VALUE_TEXT","rating":"RATING_VALUE_ALL","language":"LANGUAGE \
...

result = subprocess.run(
    curl_cmd,
    shell=True,
    capture_output=True,
    text=True
)

print("STDOUT:")
print(result.stdout)

print("STDERR:")
print(result.stderr)
```

Figure 3.4: Data Collection and Extraction Method

The research data was gathered using a secondary data extraction method through extracting publicly available customer reviews on the Traveloka online booking platform. The data collection process involved utilizing web data extraction techniques to send requests to the system's API. Each request included important parameters to retrieve and the pagination location had to be clearly defined. Thanks to the pagination mechanism, data was collected sequentially in groups of reviews until the system no longer provided new content.

The returned data from the system was stored in raw format and then transformed into a tabular structure to facilitate processing and analysis. The collected information included review IDs, scores, review content, posting timestamps, hotel responses, and some additional descriptive information. Only reviews with complete scores and comment content, publicly displayed on the platform, were selected for inclusion in the research dataset.

After collection, the data underwent a thorough cleaning process to eliminate missing values, duplicates, or inappropriate entries, while also standardizing the format to ensure consistency. The entire data collection process strictly utilized publicly available information, avoiding sensitive personal data and excluding fields that could identify individuals. The data collection and processing were carried out to serve the research purpose and comply with research ethics principles.

3.3. Data Preprocessing

3.3.1. Removing Irrelevant Variables

After collecting customer reviews, the research team performed a variable selection process to retain only the features necessary for analysis. First, the Pandas library was utilized to facilitate data manipulation and processing in tabular format. The dataset was then loaded, and a list named **cols_to_drop** was created to store the names of irrelevant variables. Subsequently, the **drop()** function provided by Pandas was applied to remove these columns from the DataFrame. The cleaned dataset, after eliminating unnecessary variables, was stored in a new variable named **df_dropped**.

```
import pandas as pd
cols_to_drop = [
    "STT",
    "productType",
    "reviewer",
    "reviewTimestamp",
    "reviewMaxScore",
    "reviewOriginalText",
    "translationStatus",
    "translationProviderLabelUrl",
    "reviewMedia",
    "reviewReply",
    "reviewAdditionalInformation",
    "subRatingInformations",
    "reactionSummary",
    "reviewOrigin",
    "reviewProvider",
    "reviewOriginLabel",
    "reviewIconUrl"
]
df_dropped = df.drop(columns=[c for c in cols_to_drop if c in
df.columns])
display(df_dropped.head())
```

The retained variables include **reviewId**, **reviewScore**, **reviewContentText**, which were then saved into a new Excel file using the **to_excel()** function.

	reviewId	ReviewScore	reviewContentText
0	1856073066124162992	10.0	nhân viên rất nhiệt tình, phòng sạch sẽ, buffe...
1	1855265970105368446	9.4	Chăm sóc khách hàng thân thiện và chuyên nghiệp!
2	1855075484958738747	10.0	Nhân viên thân thiện và nhiệt tình
3	1854849274851903803	10.0	Phòng đẹp và view đẹp, giường êm mát, tiện ích...
4	1854529681735302882	10.0	Một địa điểm tuyệt vời để đi du lịch. Mọi thứ ...

Figure 3.5 : Variables After Data Cleaning

3.3.2. Text Translation

Since the collected review data were written in multiple languages, the research team translated all review content into Vietnamese to ensure language consistency before applying analytical models. The team utilized the **deep_translator** library along with the **Google Translate** service to automatically perform text translation.

Specifically, a function named **translate_text()** was implemented to translate each review in the **reviewContentText** column. First, the function checks whether the input value is missing by using the *pd.isna()* function from the Pandas library, which is designed to detect null values (NaN). If the text is empty, the function returns the original value to avoid errors during the translation process. If the input text is valid, the program uses **GoogleTranslator** to perform the translation. The system automatically detects the source language (**source = “auto”**) and translates it into Vietnamese (**target = “vi”**).

In addition, to ensure robustness and stability during the translation process, a **try–except** structure is implemented to handle potential errors. The **try** block is responsible for executing the translation process. In case an error occurs, the program switches to the **except** block to handle the exception. At this stage, an error message is printed along with the corresponding problematic text. Meanwhile, the original text is returned to ensure that the remaining data processing workflow continues without interruption.

```
def translate_if_needed(text):
    try:
        if pd.isna(text) or str(text).strip() == "":
            return ""
        text = str(text)
        # detect language
        lang = detect(text)

        if lang != "vi":
            return GoogleTranslator(source='auto',
target='vi').translate(text)
        else:
            return text
    except:
        return text

df["review_vi"] = df["reviewContentText"].apply(translate_if_needed)
```

After completing the translation process, the review content converted into Vietnamese was stored in a new column named **review_vi**.

3.3.3. Text Normalization

Customer review content is often written in various formats, including uppercase and lowercase letters, special characters, and emojis. Therefore, the text data were normalized by converting all content to lowercase, removing emojis, special characters, and redundant whitespace in order to reduce noise and prevent distortion in the analysis results. To achieve this, the research team utilized the **emoji** library to remove emojis from the text and the **re (Regular Expressions)** library to perform string processing tasks and eliminate special characters. First, duplicate reviews were removed using the **drop_duplicates()** function in Pandas, with the parameter **subset = ["review_vi"]**, indicating that duplication was checked based on the review content column.

```
print(f"Original DataFrame shape: {df.shape}")
initial_rows = df.shape[0]
df.drop_duplicates(subset=['review_vi'], inplace=True)
duplicate_rows_removed = initial_rows - df.shape[0]
print(f"Number of duplicate rows removed: {duplicate_rows_removed}")
print(f"New DataFrame shape after removing duplicates: {df.shape}")
```

After removing duplicate data, the research team re-examined the dataset size. The results showed that 9 duplicate records were successfully eliminated, reducing the DataFrame to 2,094 rows while maintaining the original 4 columns.

Original DataFrame shape: (2103, 4)
Number of duplicate rows removed: 9
New DataFrame shape after removing duplicates: (2094, 4)

Figure 3.6: Results After Removing Duplicate Data

Next, the research team constructed a dictionary named **teencode_dict**, which contains pairs of commonly used abbreviations or “teencode” terms and their corresponding standard forms. For example, terms such as “ko”, “k”, and “hok” are converted to “không”, “dc” is replaced with “được”, and “nv” is replaced with “nhân viên”. In addition, the team developed a function called **normalize_text()** to standardize text containing “teencode”. Specifically, within this function, the text is first split into a list of individual words using the **split()** method. Then, a list comprehension combined

with the `teencode_dict` is applied to replace each word in the text. If a word exists in the dictionary, it is converted to its corresponding standard form; otherwise, it remains unchanged using the `get(word, word)` method. Finally, the normalized words are joined back into a complete string using a space separator through “`”`.`join(words)` and the processed text is returned.

```
def normalize_teencode(text):
    words = text.split()
    words = [teen_dict.get(word, word) for word in words]
    return " ".join(words)
```

In addition, the research team developed several supporting functions, including `normalize_unicode()` for Unicode standardization and `convert_emoji()` for emoji conversion. Specifically, the `normalize_unicode()` function applies the `unicodedata.normalize` method to convert text into the NFC (Normalization Form C) format. This form combines characters and diacritics into a single composed character, thereby ensuring consistency in text representation. Meanwhile, the `convert_emoji()` function is designed to transform emojis in the text into their textual descriptions using the `emoji.demojize()` function. This conversion helps preserve the emotional information conveyed by emojis in a format that can be effectively processed by NLP models, rather than being removed or introducing noise into the analysis.

```
def normalize_unicode(text):
    return unicodedata.normalize("NFC", str(text))
def convert_emoji(text):
    return emoji.demojize(text)
```

In addition, the research team developed a function named `remove_repeated_chars()` to eliminate consecutively repeated characters in the text, as well as a `clean_text()` function to perform comprehensive text cleaning. Specifically, the `remove_repeated_chars()` function applies a regular expression using `re.sub(r'(\1+)', r'\1', text)` to replace sequences of repeated characters with a single character. This step helps normalize elongated words, thereby reducing noise and limiting unnecessary vocabulary expansion within the dataset. The `clean_text()` function performs a series of text preprocessing steps. First, all text is converted to lowercase using the `lower()` method to ensure consistency. Next, URLs are removed using the regular expression `r"http\S+"`. The function then eliminates non-alphabetic

characters, including special characters and numbers, by applying the pattern `r“[^a-zA-ZÀ-ỹ\s]”`, while still preserving Vietnamese characters with diacritics. Finally, excessive whitespace is normalized to a single space using `r“\s+”`, and leading and trailing spaces are removed using the `strip()` method.

```
def remove_repeated_chars(text):
    text = re.sub(r'(\.)\1+', r'\1', text)
    return text

def clean_text(text):
    text = text.lower()
    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"[^a-zA-ZÀ-ỹ\s]", " ", text)
    text = re.sub(r"\s+", " ", text).strip()
    return text
```

Next, a function named `preprocess_text()` was developed to integrate all the previously defined functions into a unified pipeline, enabling a systematic execution of the text preprocessing steps.

```
def preprocess_text(text):
    text = normalize_unicode(text)
    text = convert_emoji(text)
    text = clean_text(text)
    text = normalize_teencode(text)
    text = remove_repeated_chars(text)
    return text
```

After completion, the `preprocess_text()` function is applied to the `review_vi` column to generate a new column named `review_clean`. Subsequently, the entire processed dataset is exported to a new Excel file to support subsequent analysis steps.

3.3.4. Tokenization and Stopword Removal

The research team utilized the **Scikit-learn**, **NLTK**, and **Underthesea** libraries to perform tokenization and stopwords removal on the text data. Initially, the **punkt** dataset for sentence and word tokenization, along with the stopwords list from the **NLTK** library, was downloaded. In addition, a custom Vietnamese stopwords list named **vietnamese_stopwords_list** was constructed. This list includes commonly occurring words that carry little semantic value for content analysis, such as “*là*”, “*và*”, “*của*”,

“*như*”, and “*vậy*”. Since the default stopwords list provided by NLTK is primarily designed for English and is not well-suited for Vietnamese text, the team developed an additional Vietnamese-specific stopwords list to improve the effectiveness of the preprocessing stage.

```
import nltk
import pandas as pd
import re
from underthesea import word_tokenize

nltk.download('punkt')
nltk.download('stopwords')

vietnamese_stopwords_list = [ "là", "mà", "thì", "và", "của", "này", "ấy",
                              "nhi", "nhé", "cho", "để", "cái", "con", "những", "tại", "với", "vậy",
                              "a", "còn", "có", "khi", "không", "là", "lại", "và", "với", "đã", "đang",
                              "được", "đi", "bị", "bởi", "cả", "các", "cái", "chỉ", "chiếc", "cho",
                              "chứ", "chợt", "chua", "chuyện", "dành", "do", "dưới", "đúng", "hết",
                              "hay", "kể", "khiến", "làm", "mà", "mỗi", "một", "muốn", "nếu", "như",
                              "nhưng", "những", "phải", "qua", "ra", "rất", "rồi", "sau", "sẽ", "tù",
                              "tùng", "trên", "trong", "trước", "vẫn", "vào", "vậy", "việc", "vì",
                              "vừa", "về"]
```

Next, the team developed a function named **preprocess_vietnamese_text()** with two parameters: `text`, which contains the input text to be processed, and `stopwords_set`, which represents the set of stopwords to be removed. First, the function checks whether the input data is a string. If the input is not a string, the function returns an empty string to prevent errors during processing. Next, the **word_tokenize()** function is applied to segment the text into individual tokens, with the parameter **format = “text”**, which ensures that the output is returned as a string where compound words are joined using underscores. For example, the phrase “khách sạn” is transformed into “khách_sạn”, allowing the model to recognize it as a single semantic unit rather than two separate words.

The resulting string is then split into a list of tokens using the **split()** function based on whitespace delimiters. Since tokens may contain underscores, the stopwords list is also normalized to match this format, ensuring consistency between tokens and stopwords. After obtaining both the token list and the standardized stopwords list,

stopwords are removed by comparing each token against the predefined stopwords set. Tokens that appear in the stopwords list are filtered out, retaining only semantically meaningful words. Finally, the remaining tokens are joined back into a new text string using the `join()` function. The `preprocess_vietnamese_text()` function is then applied to the entire `review_clean` column. The results are stored in a new column named `processed_content`, and the complete dataset is exported to a new Excel file to support the subsequent model development phase.

```
def preprocess_vietnamese_text(text, stopwords_set):
    if not isinstance(text, str):
        return ""

    text_tokenized = word_tokenize(text, format="text")
    tokens = text_tokenized.split()
    stopwords_set_underthesea_format = set([w.replace(' ', '_') for w in
stopwords_set])
    filtered_tokens = [word for word in tokens if word not in
stopwords_set_underthesea_format]
    return " ".join(filtered_tokens)

stop_words_underthesea_format = set([w.replace(' ', '_') for w in
vietnamese_stopwords_list])

df['processed_content'] = df['review_clean'].apply(
    lambda x: preprocess_vietnamese_text(x,
stop_words_underthesea_format)
)
print(df[['review_clean', 'processed_content']].head())
df.to_excel("PCCV_PTDLW_stopwords_removed.xlsx", index=False)
```

The results indicate that the tokenization and stopwords removal process was successfully carried out. The processed text became more concise and focused on the key terms that convey the main semantic meaning.

	review_clean \
0	nhân viên rất nhiệt tình phòng sạch sẽ bufet n...
1	chăm sóc khách hàng thân thiện và chuyên nghiệp
2	nhân viên thân thiện và nhiệt tình
3	phòng đẹp và view đẹp giường êm mát tiện ích t...
4	một địa điểm tuyệt vời để đi du lịch mọi thứ đ...

	processed_content
0	nhân_viên nhiệt_tình phòng sạch_sẽ bufet nhiều...
1	chăm_sóc khách_hàng thân_thiện chuyên_nghiệp
2	nhân_viên thân_thiện nhiệt_tình
3	phòng đẹp view đẹp giường êm_mát tiện_ích tốt ...
4	địa_điểm tuyệt_vời du_lịch mọi thứ đều tốt tất...

Figure 3.7: Results After Tokenization and Stopword Removal

3.4. Label Creation and Handling Imbalance

3.4.1. Label Creation (2 classes)

In essence, the dependent variable is influenced not only by ratio-scale variables but also by categorical variables. However, many Machine Learning algorithms and most Deep Learning architectures share an inherent limitation: they cannot directly process raw text data.

Modern language processing models are largely grounded in linear algebra and vector space representations to capture semantic meaning, whereas raw text consists merely of discrete character sequences that lack intrinsic numerical value. Therefore, such data must be transformed into numerical representations to enable computational operations.

This challenge is further complicated by the inherent ambiguity of natural language at multiple levels, including morphology, syntax, and deep semantics.

For instance, the sentence “*I made her duck*” is structurally simple yet can be interpreted in at least five distinct ways depending on context:

- I cooked waterfowl for her.
- I cooked waterfowl belonging to her.
- I created the (plaster?) duck she owns.

- I caused her to quickly lower her head or body.
- I waved my magic wand and turned her into a duck.

If text remains in its raw form, machine learning models are incapable of identifying the correct linguistic structure or resolving such ambiguities. Therefore, encoding techniques that transform categorical variables into numerical representations are essential.

*** Label Encoding**

One of the fundamental techniques for this transformation is Label Encoding.

In Label Encoding, each categorical label (string) is mapped to an integer value. This technique is typically applied to ordinal categorical variables, where each class is assigned an integer starting from 0.

However, a major limitation of Label Encoding is that when applied to nominal categorical variables, it unintentionally introduces artificial ordering and distance relationships that do not originally exist.

Formally, the Label Encoding function f_L maps each class c_i to an integer in the range $\{0,1,...,C-1\}$

$$CV : \begin{bmatrix} c_1 \\ c_2 \\ \dots \\ c_i \\ \dots \\ c_N \end{bmatrix}_{N \times 1} \xrightarrow{f_L} NV : \begin{bmatrix} 0 \\ 1 \\ \dots \\ i-1 \\ \dots \\ N-1 \end{bmatrix}_{N \times 1}$$

where:

- $S = \{S_1, S_2, \dots, S_C\}$ denotes the original set of labels
- $y_i \in \{0,1, \dots, C-1\}$ represents the encoded target values
- f_L is a bijective mapping ensuring a one-to-one correspondence between labels and integers

**** Application***

The process begins by constructing a function, `label_sentiment()`, to map review scores into two sentiment classes. Specifically, if the `ReviewScore` is greater than or equal to 9, it is assigned the label “positive”; otherwise, if it is less than 9, it is labeled as “negative.” This function is then applied to the entire `ReviewScore` column to generate a new column named `sentiment` using the following statement:

```
def label_review(score):  
    if score >= 9:  
        return 1  
    else:  
        return 0  
  
df["label"] = df["ReviewScore"].apply(label_review)
```

After obtaining the labels in textual form, the Label Encoding method is applied to convert these labels into numerical representations.

**** Encode label***

Specifically, a `LabelEncoder()` object is instantiated from the `scikit-learn` library to perform the encoding of categorical values, using the following statement:

```
le = LabelEncoder()
```

Here, the variable `le` serves as a reference name for the instantiated object to facilitate subsequent usage. The `fit_transform(df["sentiment"])` method is then applied to the `sentiment` column, where the `fit` step identifies all unique labels (including *positive* and *negative*), and the `transform` step maps each label to a corresponding integer value. The encoding is automatically assigned in alphabetical order, resulting in the mapping: *negative* → 0 and *positive* → 1.

```
df["label"] = le.fit_transform(df["sentiment"])
```

The result of this process is stored in a new column named `label`, which contains the numerical representations of the original categorical labels, thereby making the dataset compatible with machine learning models.

3.4.2. Computing Class Weights

```
class_weights_np = compute_class_weight(  
    class_weight='balanced',  
    classes=classes,  
    y=train_df["label"]  
)  
class_weights_tensor = torch.tensor(class_weights_np,  
dtype=torch.float32)
```

During the model training process, one of the critical challenges that must be addressed is class imbalance. In sentiment classification tasks, the number of samples across classes such as positive, neutral, and negative is often uneven, which may cause the model to become biased toward the majority class.

To mitigate this issue, a class weighting approach is employed. The class weights are computed using the `compute_class_weight` function from the `sklearn.utils` library. With the parameter `class_weight="balanced"`, this function automatically assigns weights based on the frequency of each class in the dataset. Specifically, classes with fewer samples are assigned higher weights, while majority classes receive lower weights. This mechanism encourages the model to pay more attention to minority classes during training.

After computation, the class weights are converted into tensor format using PyTorch, allowing them to be directly integrated into the loss function during the training process, particularly in the fine-tuning of the PhoBERT model. Incorporating class weights into the loss function helps balance the contribution of each class, thereby improving the overall performance of the model, especially the F1-score for underrepresented classes.

3.5. Text Feature Extraction Techniques

3.5.1. Data Representation using TF-IDF (for SVM)

** Rationale for Method Selection*

Managing repositories of unstructured and semi-structured documents has become increasingly challenging due to the exponential growth in both the volume and number of documents. Identifying groups of similar documents has therefore become a

crucial task in order to eliminate duplicate documents, detect plagiarism, and support search engines in collecting and organizing data more effectively.

However, clustering techniques generally require data to be represented in a structured format. Therefore, textual documents must be transformed into a vector space model, typically represented as a term-document matrix. In practice, extracting all terms from documents can generate an extremely large vocabulary, which leads to the curse of dimensionality, significantly reducing the efficiency of machine learning algorithms.

The TF-IDF (Term Frequency-Inverse Document Frequency) method is selected to address this issue due to two main advantages. First, it helps remove common words that appear frequently across documents but carry little discriminative meaning. Second, it highlights and retains the most informative features (keywords), accurately reflecting the importance of a term within a specific document collection.

**** Role in the Machine Learning Model***

Machine learning algorithms such as clustering and classification operate on numerically structured data. Therefore, TF-IDF is employed as a bridge to transform unstructured textual data into a term-document matrix. In this representation, documents correspond to the rows, while terms represent the columns, with each cell containing the corresponding TF-IDF weight.

By combining stopwords removal with TF-IDF weighting, the machine learning model is trained on input features that possess stronger discriminative power. This preprocessing step enables the model to focus on the most informative terms, thereby improving the accuracy and effectiveness of clustering or classification results in the analysis of review data.

**** Applying TF-IDF to the Real-World Dataset***

After completing the text preprocessing stage, including stopwords removal and data normalization, the dataset must be labeled for classification and the textual content needs to be transformed into a vector space representation.

This process is implemented using the Python programming language, with the support of libraries such as pandas and scikit-learn. The research dataset consists of 2,094 preprocessed reviews, where each review is assigned a sentiment label belonging

to one of three categories: positive, neutral, or negative, based on the corresponding ReviewScore value.

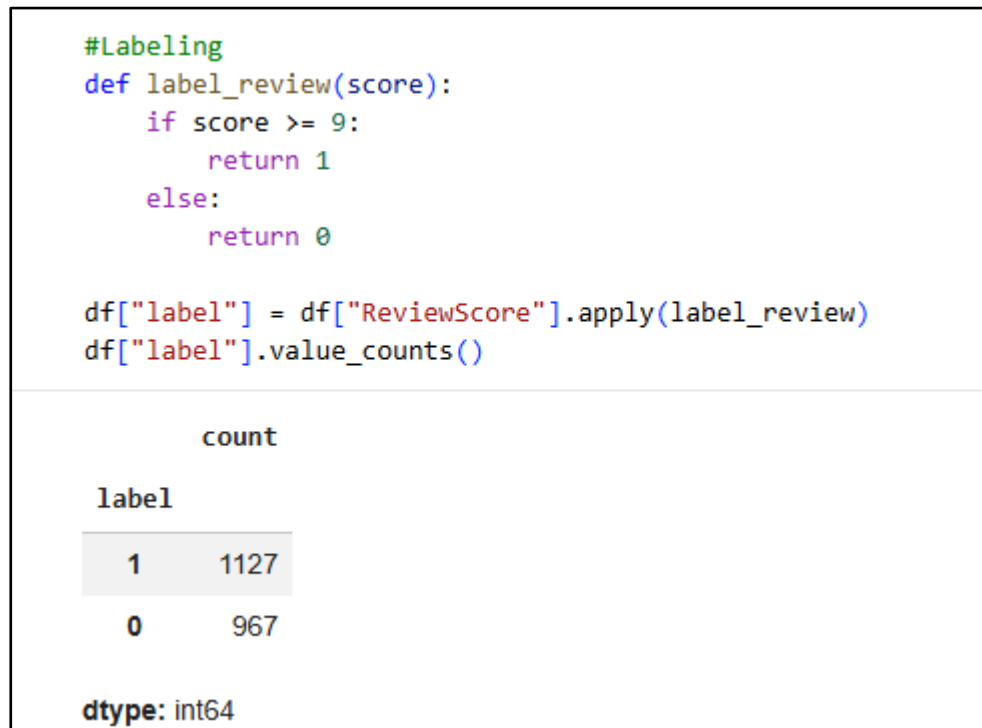


Figure 3.8: Rule-Based Sentiment Classification from Review Scores

Next, to perform the text vectorization process, the TfidfVectorizer model was initialized with carefully adjusted parameters to optimize the feature extraction process.

```
from sklearn.feature_extraction.text import TfidfVectorizer

vectorizer = TfidfVectorizer(
    min_df=1,
    max_df=0.95,
    ngram_range=(1,2)
)

# Fill NaN values with empty strings before vectorizing
X = vectorizer.fit_transform(df["processed_content"].fillna(""))
y = df["label"]

print("TF-IDF shape:", X.shape)
```

3.5.2. Data Representation using PhoBERT Tokenizer

```
tokenizer = AutoTokenizer.from_pretrained("vinai/phobert-base")

def tokenize_function(examples):
    return tokenizer(
        examples["review_clean"],
        padding="max_length",
        truncation=True,
        max_length=256
    )
```

In the next step, tokenization is performed to convert textual data into a numerical format that the model can process. Specifically, a function named `tokenize_function` is implemented to apply the tokenizer to the `review_clean` column, which contains the text that has already been preprocessed. The use of a tokenizer transforms each sentence into a sequence of tokens and maps these tokens into their corresponding numerical indices in the model's vocabulary.

Within the `tokenizer()` function, the process involves not only token splitting but also the encoding of text into numerical vectors. The parameter `padding="max_length"` is used to ensure that all sentences have the same length by adding padding tokens to shorter sequences. This is particularly important during batch training, as deep learning models require input data with consistent dimensions. At the same time, the parameter `truncation=True` allows sentences that exceed the maximum allowed length to be truncated, helping control data size and reduce computational cost.

The value `max_length=256` is selected as the maximum length for each token sequence. This value is commonly used in practice because it balances the need to preserve most of the sentence information while maintaining computational efficiency. Choosing this length also helps avoid excessively long sequences that could consume more memory or slow down the training process.

As a result of the tokenization step, each document in the `review_clean` column is converted into a fixed-length numerical vector containing standardized token indices. This transformation makes the dataset suitable for input into deep learning models. Although this step serves as an intermediate stage, it plays a crucial role, as the quality of the tokenization process directly affects the model's ability to capture semantic meaning and ultimately influences its prediction performance.

CHAPTER 4: MODEL DEVELOPMENT AND EVALUATION

4.1. Sentiment Classification Model Development

4.1.1. Initialization and Training of the Support Vector Machine (SVM)

Model

From a theoretical perspective, Support Vector Machine (SVM) aims to identify an optimal hyperplane that maximizes the margin between classes. The model focuses primarily on data points located near the decision boundary, known as support vectors. This characteristic enhances the model's generalization capability, particularly in scenarios involving noisy data or observations close to class boundaries.

The model is implemented using LinearSVC to directly optimize a linear classification problem instead of employing nonlinear kernels. This approach is based on the assumption that the separation structure between sentiment classes can be effectively modeled using a linear hyperplane in the TF-IDF feature space. The use of a linear kernel helps reduce the risk of overfitting and ensures scalability when the feature dimensionality is high. Additionally, the SVM is configured with a soft margin mechanism, allowing for a certain level of misclassification to improve generalization performance. The regularization parameter C is kept at its default value, representing a standard trade-off between maximizing the margin and minimizing training errors. This configuration maintains model simplicity during the initial training phase while enabling fair comparison with other linear models.

```
from sklearn.svm import LinearSVC

svm_model = LinearSVC(class_weight="balanced")
svm_model.fit(X_train, y_train)
```

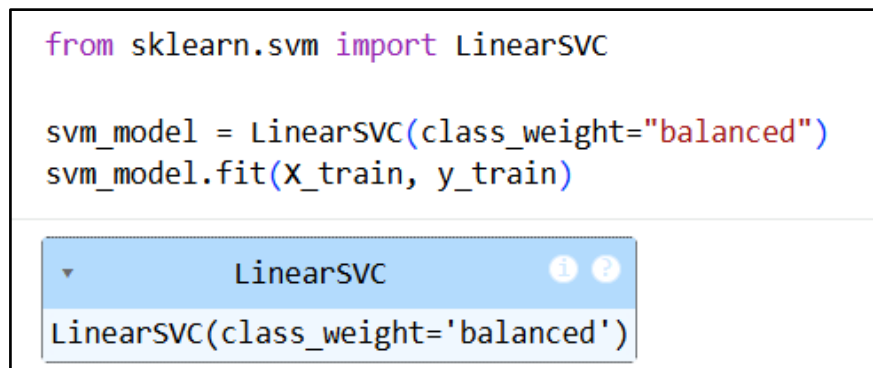


Figure 4.1: SVM Model Initialization and Training Setup

The model incorporates the parameter `class_weight = "balanced"` to mitigate the impact of class imbalance. This mechanism automatically adjusts class weights inversely proportional to class frequencies, thereby increasing the importance of minority classes.

The core training process is executed through the `.fit(X_train, y_train)` method. The `fit()` function of `LinearSVC` processes the training data to solve the underlying geometric optimization problem. Specifically, it computes the optimal hyperplane by maximizing the margin while satisfying constraints related to class balance and soft margin regularization.

After training on the training set, the model is evaluated on the test set using standard performance metrics, including Precision, Recall, F1-score, and Accuracy.

The SVM model serves as an important baseline model to validate the stability of classification results within the TF-IDF feature space and to support insights into the structural characteristics of sentiment data. The model development process follows a structured approach, including selecting an appropriate linear architecture, controlling model complexity, and addressing class imbalance during training. This ensures that the model is theoretically grounded, scalable, and well-suited for text classification tasks.

4.1.2. Initialization and Fine-tuning of the PhoBERT Model

In the sentiment classification task using hotel review data, PhoBERT is selected as the primary model due to its strong capability in understanding Vietnamese semantics through the Transformer architecture. The training and fine-tuning process is designed as a structured pipeline to optimize performance, particularly under conditions of limited data and class imbalance.

```
class WeightedTrainer(Trainer):
    def compute_loss(self, model, inputs, return_outputs=False,
**kwargs):
        labels = inputs.pop("labels")
        outputs = model(**inputs)
        logits = outputs.get("logits")

        if hasattr(self, 'class_weights') and self.class_weights is not
None:
```



```

        loss_fct =
CrossEntropyLoss(weight=self.class_weights.to(logits.device))
    else:
        loss_fct = CrossEntropyLoss()
        loss = loss_fct(logits.view(-1, self.model.config.num_labels),
labels.view(-1))
    return (loss, outputs) if return_outputs else loss

```

To incorporate class weights into the training process, a custom training class (CustomTrainer) is implemented by extending the Trainer class from the Hugging Face Transformers library. This allows direct modification of the loss computation process to effectively handle class imbalance.

Specifically, the `compute_loss` method is overridden to utilize a weighted `CrossEntropyLoss` function. During training, the input data consists of tokenized features and corresponding labels. The model performs a forward pass to produce logits, which are then fed into the loss function along with the true labels.

The key difference lies in initializing the `CrossEntropyLoss` with the parameter `weight=class_weights`, where class weights are precomputed based on the dataset distribution. These weights are also transferred to the same computational device (CPU or GPU) as the model using `.to(model.device)` to ensure consistency. Additionally, the inclusion of `**kwargs` in the function definition improves flexibility and compatibility with different versions of the Trainer class, preventing potential errors when unexpected arguments are passed.

This approach increases the penalty for misclassification of minority classes, encouraging the model to better learn from underrepresented samples.

```

training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=1e-5,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    num_train_epochs=4,
    weight_decay=0.01,
    logging_dir="./logs",
    eval_strategy="epoch",
    save_strategy="no",
    fp16=True)

```

During fine-tuning, training parameters are defined using the `TrainingArguments` class from the Hugging Face Transformers library. Output and logging directories are specified via `output_dir` and `logging_dir` to facilitate monitoring and result storage.

The learning rate is set to $1e-5$, which is commonly used for fine-tuning Transformer-based models like PhoBERT, ensuring stable convergence without overwriting pretrained knowledge. The batch size for both training and evaluation is set to 8 samples per step, which is suitable for limited hardware resources and helps prevent memory overflow.

The number of training epochs is limited to 4 (`num_train_epochs=4`) to ensure sufficient learning while avoiding overfitting, especially given the relatively small dataset size. Additionally, `weight_decay=0.01` is applied as a regularization technique to control model complexity and reduce overfitting.

The evaluation strategy is set to `eval_strategy="epoch"`, meaning that the model is evaluated at the end of each epoch. This enables monitoring performance across training stages and supports model selection. Meanwhile, `save_strategy="no"` disables checkpoint saving to reduce training time and storage usage.

```
fold_accuracies = []
fold_f1s = []
fold_recalls = []

for fold, (train_idx, val_idx) in enumerate(skf.split(df, df["label"])):
    print(f"\n===== Fold: {fold} =====")
    train_df = df.iloc[train_idx].copy()
    val_df = df.iloc[val_idx].copy()
    train_df['review_clean'] = train_df['review_clean'].fillna('')
    val_df['review_clean'] = val_df['review_clean'].fillna('')
    train_dataset =
Dataset.from_pandas(train_df[['review_clean', 'label']])
    val_dataset = Dataset.from_pandas(val_df[['review_clean', 'label']])
    train_dataset = train_dataset.map(tokenize_function, batched=True)
    val_dataset = val_dataset.map(tokenize_function, batched=True)
    train_dataset.set_format(type="torch",
columns=["input_ids", "attention_mask", "label"])
    val_dataset.set_format(type="torch",
columns=["input_ids", "attention_mask", "label"])
```

```

model = AutoModelForSequenceClassification.from_pretrained(
    "vinai/phobert-base",
    num_labels=len(np.unique(df["label"]))
)

```

The dataset is divided into multiple folds, where each fold serves as a validation set while the remaining folds are used for training. Stratification ensures that the label distribution in each fold reflects that of the overall dataset.

In each iteration, training and validation subsets are extracted using index pairs (train_idx, val_idx) and copied to avoid modifying the original data. Missing values in the text column (review_clean) are handled by replacing them with empty strings to ensure consistency during tokenization.

The data is then converted into a format compatible with the Hugging Face library using Dataset.from_pandas. Tokenization is performed in batches to improve computational efficiency, producing features such as input_ids and attention_mask as model inputs.

The model used in this study is PhoBERT, specifically the pretrained version “vinai/phobert-base”, which has been trained on large-scale Vietnamese corpora. Leveraging this pretrained model significantly enhances performance in sentiment analysis tasks. The number of output classes (num_labels) is determined based on the unique values of the target label.

The variable best_f1 is initialized to 0, and the optimal threshold best_t is initially set to 0.5. True labels for the validation set are obtained from val_df[“label”].values.

A loop is then performed over threshold values [0.4, 0.5, 0.6, 0.7]. For each threshold, predicted probabilities for the positive class (probs[:,1]) are converted into binary labels using (probs[:,1] > t).astype(int). The F1-score is computed using f1_score with average=“binary”. If the current F1-score exceeds the previous best, both best_f1 and best_t are updated accordingly.

After determining the optimal threshold, final predictions are generated and stored in the original DataFrame (df) at the corresponding validation indices (val_idx) under the column “pred_label”.

Evaluation metrics including Accuracy, F1-score, and Recall are computed and stored in lists (fold_accuracies, fold_f1s, fold_recalls) for aggregation across folds.

Finally, results for each fold are printed, and the overall performance is summarized by calculating the mean Accuracy and F1-score using np.mean. To execute this pipeline, it is required that variables such as probs, val_df, df, val_idx, and the result storage lists are properly initialized beforehand.

4.2. Evaluation and Validation Methods

4.2.1. Stratified K-Fold Cross Validation

To ensure reliable evaluation and robust measurement of the model's generalization capability, this study adopts the Stratified K-Fold Cross Validation technique for training and validating the PhoBERT model. K-Fold Cross Validation is a widely used evaluation method in machine learning. In this approach, the dataset is randomly divided into K equal subsets (folds). The model is trained on (K - 1) folds and validated on the remaining fold, and this process is repeated K times.

This method enables efficient utilization of the dataset and allows evaluation across multiple validation sets, thereby reducing estimation bias and minimizing the risk of misleading performance evaluation due to a particular data split (Kohavi, 1995). However, in classification tasks with imbalanced data, random splitting may distort the class distribution across folds, leading to biased model evaluation.

To address this issue, the Stratified K-Fold variant is employed, ensuring that each fold preserves the same class distribution as the original dataset. According to Pedregosa et al.(2011), this approach is a standard practice in modern machine learning libraries to improve representativeness and model stability.

```
skf = StratifiedKFold(  
    n_splits=5,  
    shuffle=True,  
    random_state=42)  
scores = []
```

Following Kohavi (1995), K is typically set to 5 or 10 in empirical studies. Given the dataset size of approximately 2000 observations, this study adopts K = 5.

The parameters `shuffle = True` and `random_state = 42` are used to randomize the data while maintaining reproducibility, thereby minimizing the impact of the original data order.

In each iteration, one fold is used as the validation set, while the remaining folds are used for training. Evaluation metrics such as Accuracy, Precision, Recall, and F1-score are computed for each iteration, and their average values are used to represent overall model performance.

The application of Stratified K-Fold not only enables comprehensive evaluation across multiple data splits but also enhances the stability and generalization capability of the PhoBERT model in Vietnamese sentiment analysis. This step serves as a crucial foundation before proceeding to more advanced tasks such as negative review filtering, topic modeling, and aspect extraction.

4.2.2. Train - Test Split using Scikit-learn

In machine learning, the Train - Test Split method is a fundamental technique used to evaluate a model's generalization performance on unseen data. According to Hastie et al.(2009), datasets are typically divided into a training set and a testing set for performance evaluation.

In classification problems involving multiple classes or class imbalance, maintaining consistent class distribution across subsets is essential to ensure representativeness. For SVM, as proposed by Cortes & Vapnik (1995), the learned decision boundary is highly dependent on the data structure. Therefore, imbalanced class distributions may lead to biased learning outcomes.

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, df["label"],
    test_size=0.2,
    random_state=42,
    stratify=df["label"]
)
```

In this study, the dataset is split into training and testing sets using the `train_test_split` function from the Scikit-learn library, with a ratio of 80% - 20%. The parameter `stratify = df["label"]` is applied to preserve class distribution across subsets.

With this configuration, the training set (1600 samples) is sufficiently large for the SVM model to learn decision boundaries, while the testing set (400 samples) remains adequate for reliable evaluation. The train-test split serves as an initial evaluation step to quickly assess model performance on unseen data. However, since it relies on a single split, it may introduce estimation bias.

4.2.3. Evaluation Metrics

To assess model performance comprehensively and objectively, this study employs widely used evaluation metrics for classification tasks.

- **Accuracy** measures the proportion of correctly predicted instances over the total number of observations. It provides a general indication of model performance and is widely used due to its simplicity and interpretability.
- **Precision** answers the question: “Among the instances predicted as positive, how many are truly positive?” It measures the correctness of positive predictions.
- **Recall** evaluates the model’s ability to identify all relevant instances of a given class. A high recall indicates that the model successfully captures most true positive cases, which is particularly important in detecting negative reviews that reflect service issues.
- **F1-score** is the harmonic mean of Precision and Recall, providing a balanced evaluation between correctness and completeness. It is especially useful in imbalanced datasets, where Accuracy alone may be misleading.

4.3. Comparison between SVM and PhoBERT

=== SVM ===				
	precision	recall	f1-score	support
negative	0.70	0.68	0.69	193
positive	0.73	0.76	0.75	226
accuracy			0.72	419
macro avg	0.72	0.72	0.72	419
weighted avg	0.72	0.72	0.72	419

Figure 4.2: Results from SVM

===== TỔNG KẾT SAU KHI CHẠY XONG K-FOLD =====				
Mean Accuracy: 0.7531				
Mean F1-Score: 0.7921				
Mean Recall: 0.8740				

Figure 4.3: Results from PhoBERT

A detailed analysis of model performance reveals differences in Precision, Recall, and F1-score across classes, indicating a slight imbalance in classification capability. Specifically, the positive class achieves a higher recall (0.76) compared to the negative class (0.68), suggesting that the model is more effective in identifying positive instances but may miss some negative cases. This implies that the model tends to reduce false negatives for the positive class while increasing misclassification for the negative class.

Although the overall accuracy reaches 0.72 and the macro and weighted averages are relatively consistent, there remains a noticeable gap in F1-score between classes (0.69 vs. 0.75). This indicates that the model does not fully balance Precision and Recall across the dataset.

When compared with K-Fold Cross Validation results, the average performance metrics are significantly higher, particularly Recall (0.8740). While this suggests improved coverage of positive samples, it may also indicate an increase in false positives.

The discrepancy between single test set results and K-Fold results highlights the issue of data representativeness. A single test split may not accurately reflect overall model performance, whereas K-Fold provides a more stable and reliable evaluation by averaging across multiple data partitions.

Although SVM demonstrates relatively good performance with low computational cost, PhoBERT is considered more suitable for this task. PhoBERT is capable of improving both Accuracy and F1-score while offering better scalability and effectiveness for more complex natural language processing tasks.

4.4. Negative Aspect Extraction Method

4.4.1. Extracting Negative Reviews Using the Best-Performing Model

```
df_neg = df[df["pred_label"] == 0].copy()
texts = df_neg["review_clean"].astype(str).tolist()
print(f"Số lượng đánh giá tiêu cực: {len(texts)}")
print(f"Kiểu dữ liệu phần tử đầu tiên: {type(texts[0])}")
Số lượng đánh giá tiêu cực: 734
Kiểu dữ liệu phần tử đầu tiên: <class 'str'>
```

The filtering results indicate that the model identified 734 negative reviews from the entire dataset. This represents a relatively substantial number, suggesting that the classification model successfully detected a sufficiently large group of customer feedback expressing dissatisfaction. Extracting a separate set of negative reviews enables a more focused analysis of existing issues in service quality, thereby supporting the development of appropriate improvement strategies.

In addition, the data type verification shows that each element in the text corpus is stored as a string. This confirms that the data has been properly formatted and is ready for subsequent analysis steps such as keyword extraction, frequency analysis, or the application of other natural language processing techniques. Storing the data in string format also helps prevent potential errors during the processing stage.

4.4.2. Applying the BERTopic Model for Topic Modeling

```
from bertopic import BERTopic
from sentence_transformers import SentenceTransformer

embedding_model = SentenceTransformer("paraphrase-multilingual-MiniLM-
L12-v2")

topic_model = BERTopic(
    embedding_model=embedding_model,
    min_topic_size=5,
    nr_topics="10"
)
topics, probs = topic_model.fit_transform(texts)
```

In this step, the study applies the BERTopic model to extract latent topics from the previously filtered set of negative reviews. Specifically, the model employs a multilingual language embedding model to transform textual data into semantic vectors, enabling the model to capture the contextual meaning of sentences rather than relying solely on word frequency. The use of a multilingual model is appropriate in this context, as the dataset may contain multiple languages or diverse linguistic expressions.

The model parameters are configured with **min_topic_size = 5**, meaning that each topic must contain at least five reviews. This constraint helps ensure that the identified topics are sufficiently representative and prevents the generation of overly small topics that are difficult to interpret. In addition, the parameter **nr_topics = 10** is used to limit the final number of topics to ten, allowing the analysis results to remain concise and easier to present in the study.

```
Loading weights:  0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2
Key               | Status      | |
-----+-----+-----+
embeddings.position_ids | UNEXPECTED | |

Notes:
- UNEXPECTED      :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

Figure 4.4: Embedding Model Loading Warning

During the model training process, the system displayed a notification related to loading the weights of the embedding model. Specifically, the message

“UNEXPECTED” appeared for several components such as *embeddings.position_ids*. However, according to the accompanying note, this warning can be safely ignored when the model is used for a purpose different from its original architecture. Therefore, it does not affect the functioning of the model nor introduce bias into the topic analysis results.

It can thus be concluded that the model was successfully initialized and loaded, and is ready for the topic extraction step on the negative review dataset. The warnings that appear during the model loading process are primarily technical in nature and do not significantly impact the quality of the analytical results.

```
stopwords = [
    "không", "có", "là", "mình", "thì", "rất",
    "và", "cũng", "nhưng", "được", "bị"
]

def clean_words(topic_id):
    words_scores = topic_model.get_topic(topic_id)
    words = [
        w for w, s in words_scores
        if s > 0.03 and w not in stopwords and len(w) > 2
    ]
    return words[:10]
```

In the next step of applying the BERTopic model, the study performs a cleaning and selection process for the keywords representing each topic in order to improve the interpretability of the results. Specifically, a list of stopwords is predefined, consisting of commonly occurring Vietnamese words that carry little discriminative meaning in terms of content, such as “không”, “có”, “là”, “và”, and “nhưng”. Removing these words helps reduce noise and allows the analysis to focus on keywords with higher informational value.

A function named `clean_words` is developed to extract the most relevant keywords from each topic generated by the model. First, the function retrieves the list of words and their corresponding weights for a given topic. The words are then filtered based on three main criteria: (1) only words with a weight greater than 0.03 are retained to ensure sufficient relevance to the topic; (2) words included in the predefined stopword list are removed; and (3) very short words are excluded, as they typically carry limited analytical meaning.

After applying these filtering conditions, the function returns up to 10 keywords that most strongly represent each topic. This approach ensures that the topics are characterized by clear, interpretable, and informative keywords, thereby facilitating a more effective analysis of the content within tourists' negative reviews.

```
def clean_text(text):
    text = str(text).lower()
    text = re.sub(r"\b(the|is|and|to|was)\b", "", text)
    return text

df_neg["clean_topic"] = df_neg["review_clean"].apply(clean_text)
```

Next, the study performs text normalization to improve the quality of the input data for the BERTopic model. Specifically, a function named **clean_text** is developed to process each negative review in the dataset.

First, all textual content is converted to lowercase to ensure consistency and avoid situations where the same word written in different cases is treated as separate features. Next, several common English words such as “the”, “is”, “and”, “to”, “was” are removed from the text. These words appear frequently but carry limited semantic meaning; therefore, removing them helps reduce noise and allows the model to focus on more informative keywords.

After the cleaning process, the processed data is stored in a new column named **clean_topic**. Creating this separate column helps preserve the original data while generating a normalized version that can be directly used for the topic extraction process. This preprocessing step contributes to improving the clarity of the detected topics, enabling the analysis results to more accurately reflect the issues that lead to tourists' dissatisfaction with the service experience.

4.4.3. Semantic Similarity–Based Aspect Mapping

```
from sentence_transformers import util

aspect_labels = [
    "Facilities",
    "Amenities",
    "Cleanliness",
    "Food",
    "Staff&Service"]
```

```

aspect_embeddings = embedding_model.encode(aspect_labels,
convert_to_tensor=True)

def map_topic_semantic(words):
    text = " ".join(words)
    topic_embedding = embedding_model.encode(text,
convert_to_tensor=True)
    scores = util.cos_sim(topic_embedding, aspect_embeddings)
    best_idx = scores.argmax()
    return aspect_labels[best_idx]

```

First, a set of key aspects is defined, including Facilities, Amenities, Cleanliness, Food, and Staff & Service. These aspects represent common factors that reflect guests' experiences during their stay at a hotel. Each aspect is then transformed into a semantic vector using the same language embedding model applied in the previous steps. This transformation enables the model to capture the semantic meaning of each aspect rather than treating them merely as simple keywords.

Next, for each topic, the representative keywords are combined into a short piece of text. This text is also converted into a corresponding semantic vector representation. The semantic similarity between the topic and each predefined aspect is then calculated using cosine similarity. The result is a set of similarity scores that indicate the degree of semantic closeness between the topic and each aspect.

Finally, the aspect with the highest similarity score is selected as the representative label for that topic. This approach allows topics to be automatically mapped to specific content categories based on semantic relationships, rather than relying entirely on subjective interpretation. As a result, the analysis becomes more consistent and more accurately reflects the aspects of the service experience that tourists care about or are dissatisfied with.

<pre> topic_data = [] for topic_id in topic_model.get_topics(): if topic_id == -1: continue words = clean_words(topic_id) if len(words) == 0: continue topic_data.append({ "Topic": topic_id, "Keywords": ", ".join(words), "Aspect": map_topic_semantic(words) }) df_topics = pd.DataFrame(topic_data) df_topics </pre>			
	Topic	Keywords	Aspect
0	0	phòng	Facilities
1	1	boi, nhỏ	Amenities
2	2	nhan, khách, khong, san, vien, phong, sach, tot	Amenities
3	3	khách	Facilities
4	4	check	Amenities
5	5	sáng, khách, sạn	Facilities
6	6	sáng	Amenities
7	7	sạn, khách, cấp	Facilities
8	8	món, ngon, sáng, bữa	Food
9	9	viên, nhân, hàng, tân, check, thái, cười	Staff&Service
10	10	đăng, chặt, thịt, tiền, giá, buffet, quá, xứng,...	Food
11	11	tiếng, cách, tivi, nghe, kém, mọi, netflix	Food
12	12	phòng, nội, thất, hành, như, sao, diện, xách	Facilities
13	13	biển, bãi, một, gần	Food
14	14	sạn, khách, của, rãi, chơi, dẫn	Facilities
15	15	sạn, tàu, vũng, khách, khoái, nang, mừng, phườ...	Facilities
16	16	mùi, hôi, chịu, khó, phòng, mốc, cái, gạt	Cleanliness
17	17	lạnh, máy, tắt, đồng, làm, nóng, hỏng, chỉnh, gọi	Cleanliness
18	18	wifi, yếu, tôi, vừa	Amenities
19	19	trẻ, tôi, thu, phụ, trả, phòng, việc, đến	Staff&Service
20	20	chỗ, frowning, mới, đậu, khách, giá	Facilities
21	21	biển, view, nhìn, kính, bụi, checkin, chụp, lâu	Cleanliness
22	22	lung, bed, chăn, giường, khách sạn, tiêu, nằm, ...	Facilities
23	23	dịch, xứng, chất, ích, tốt, tương, vật	Cleanliness
24	24	bạn, chồng, fusion	Amenities
25	25	phòng, grand, khoáng, thoáng, hơi	Facilities
26	26	hộp, tiếp, tầng, muối	Staff&Service

Figure 4.5: Construction of Topic DataFrame with Keywords and Mapped Aspects

In the next step, after constructing the semantic mapping function, the study applies this procedure to all topics extracted from the BERTopic model. Specifically, the program iterates through each topic, removing the noise topic (denoted as -1) as well

as topics that do not contain representative keywords. For each valid topic, the key terms are extracted using the **clean_words** function and subsequently used to determine the corresponding aspect through semantic similarity measurement.

The results show that the model identified 27 distinct topics, each represented by a set of characteristic keywords and assigned to one of the predefined aspects. The main aspects include Facilities, Amenities, Cleanliness, Food, and Staff & Service. It can be observed that the topics are not evenly distributed across these aspects.

Specifically, the Facilities aspect appears frequently, with topics related to physical infrastructure such as “phòng”, “wifi”, “biển”, and “vị trí”, reflecting issues related to physical amenities and the accommodation environment. The Cleanliness aspect also accounts for a significant proportion, with keywords such as “mùi”, “hôi”, “móc”, “sạch”, and “tắm”, indicating that cleanliness is one of the factors most frequently mentioned by tourists when expressing dissatisfaction. This suggests that hygiene conditions and room maintenance have a substantial impact on negative customer experiences.

In addition, the Amenities aspect includes topics associated with facilities such as “hồ bơi”, “ăn sáng”, or other complementary services. The Staff & Service aspect reflects issues related to staff attitudes and service procedures, with keywords such as “nhân viên”, “check”, “tiếp”, and “trả phòng”. Topics in this group indicate that certain aspects of service delivery may still fall short of customer expectations.

However, it can be observed that some topics contain keywords that are not entirely clear or still exhibit a degree of noise, which may affect the accuracy of the aspect mapping process. Furthermore, the Food aspect does not appear in the results, suggesting that negative reviews rarely mention food-related issues.

Overall, the mapping results indicate that the model has categorized the topics into specific aspects in a relatively reasonable manner, helping to clarify the key issues that lead to tourists’ dissatisfaction. This provides an important foundation for deeper analysis and for proposing solutions to improve service quality in the accommodation sector.

CHAPTER 5: RESULTS AND DISCUSSION

5.1. Sentiment Analysis Results

To gain deeper insights into the factors contributing to negative reviews, we filter reviews classified as *negative* and apply the BERTopic model to extract the main topics. The results indicate that negative customer feedback is concentrated around several prominent aspects.

Specifically, the identified topics primarily relate to room quality (e.g., small rooms, unpleasant odors, lack of cleanliness), service (e.g., staff attitude and responsiveness), and pricing (e.g., high prices relative to perceived quality). The representative keywords for each topic are visualized as bar charts, enabling clearer identification of the core issues experienced by customers.

Furthermore, a WordCloud visualization is employed to illustrate the frequency distribution of words extracted from negative reviews, providing an intuitive representation of commonly mentioned concerns.



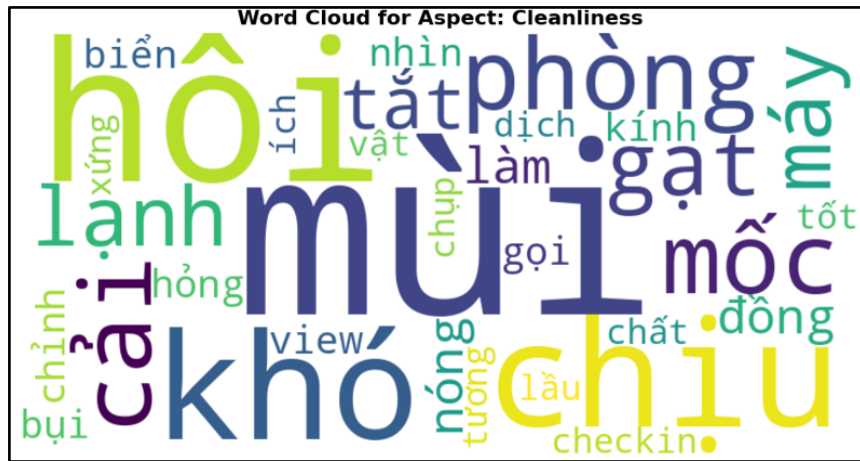


Figure 5.1: Word Cloud Visualization of Key Aspects Across Service Categories

WordCloud imagery shows that keywords such as “phòng”, “khách”, “sạn”, “sáng”, “nội thất”, “giá”, “wifi”, “mùi”, “hôi”, “nhỏ”, “yếu” appear frequently, reflecting relatively clearly both positive and negative aspects of customer experience. The combination of the BERTopic model and Word Cloud not only systematically identifies key topics but also provides a visual perspective, highlighting core issues in the accommodation experience.

Specifically, in terms of Facilities, phrases like “phòng”, “nội thất”, “sáng”, “thoáng”, “đẹp”, “giường”, “chăn” indicate that customers generally have a positive assessment of the accommodation space and basic amenities.

For Amenities, keywords like “hồ bơi nhỏ”, “wifi yếu”, “check”, “sạch”, “tốt” reflect an average level of amenity experience, with both positive and negative aspects. However, the presence of human-related phrases such as “nhân”, “khách”, “viên” indicates that the model doesn't clearly distinguish between amenities and services, leading to misinterpretation.

In the Food aspect, phrases like “delicious food,” “món ngon”, “bữa sáng”, “buffet”, “đáng tiền”, “xứng” show positive assessments of food quality. However, words like “chật”, “giá”, “quá” reflect dissatisfaction with space and cost. Notably, many words unrelated to food, such as “tivi”, “netflix”, “tiếng ồn” or “biển”, “bãi”, “gần” are still included in this group, indicating significant inaccuracies in topic classification.

For Staff & Service, phrases like “nhân viên”, “thái độ”, “cười”, “check”, “hỗ trợ” show many positive responses regarding friendliness and professionalism.

However, issues such as “phụ thu”, “trả thêm”, “trẻ em” reflect dissatisfaction with policies and operational procedures. Additionally, the appearance of words like “muỗi”, “tàng” suggests a mix of service and physical conditions.

In terms of Cleanliness, negative keywords such as “mùi”, “hôi”, “mốc”, “khó chịu”, “bụi” appear frequently, indicating a significant bottleneck. However, technical issues like “máy”, “lạnh”, “hỏng”, “chỉnh”, “tắt” or experiential elements like “view”, “biển”, “check-in” are also categorized in this group, reflecting a confusion between hygiene, amenities, and the overall experience. Furthermore, general evaluative phrases such as “dịch” (vụ) “tốt”, (chất lượng) “xứng” also appear in the Cleanliness group, indicating that the model has not clearly separated the evaluation dimensions.

Overall, the analysis results show that the customer experience is fragmented, with a mix of positive and negative elements. More importantly, the machine learning model is still not optimized, as evidenced by numerous inaccuracies in classification and semantic representation. Specifically, the model incorrectly labels aspects when comfort issues such as malfunctioning, hot, off, or difficult-to-adjust air conditioning are grouped under Cleanliness, while content related to the check-in process is mistakenly classified under Amenities.

In addition, semantic mixing occurs when different elements such as entertainment, space, and overall assessment are grouped together, reducing the consistency of the topic. Simultaneously, the model suffers from fragmentation when words with common meanings like “room” and “guest” are separated into individual topics.

Furthermore, limitations in Vietnamese preprocessing cause compound words to be separated or lose their diacritical marks, degrading semantic representation. Consequently, in some aspects, particularly regarding Food, the model fails to extract specific features and instead aggregates general terms, leading to misinterpretations. These limitations reduce the system's reliability and directly impact the validity of research conclusions.

5.2. Identification of Bottlenecks and Service Quality Gaps

Conducting an in-depth analysis of tourists’ negative reviews plays a pivotal role in enabling 4-star hotels in Vung Tau to obtain a more objective perspective on actual

service quality. Although the hotels generally receive a high proportion of positive reviews, there still exist “bottlenecks” that render the accommodation experience of a certain customer segment incomplete. Based on the topic clustering results derived from the BERTopic model, it is evident that the complained issues are not randomly distributed but rather converge into “bottleneck clusters” corresponding to specific aspects. On this premise, the study will focus on analyzing the core aspects requiring prioritized improvement and contrasting the current situation with the stringent standards of 5-star hotels, aiming to elucidate the quality gap and provide directions for future quality enhancement.

5.2.1. Service Aspects Requiring Improvement

The analysis results of the negative topic clusters indicate that customer dissatisfaction stems not from random errors or unexpected factors, but from systemic loopholes within the hotel’s service operations chain.

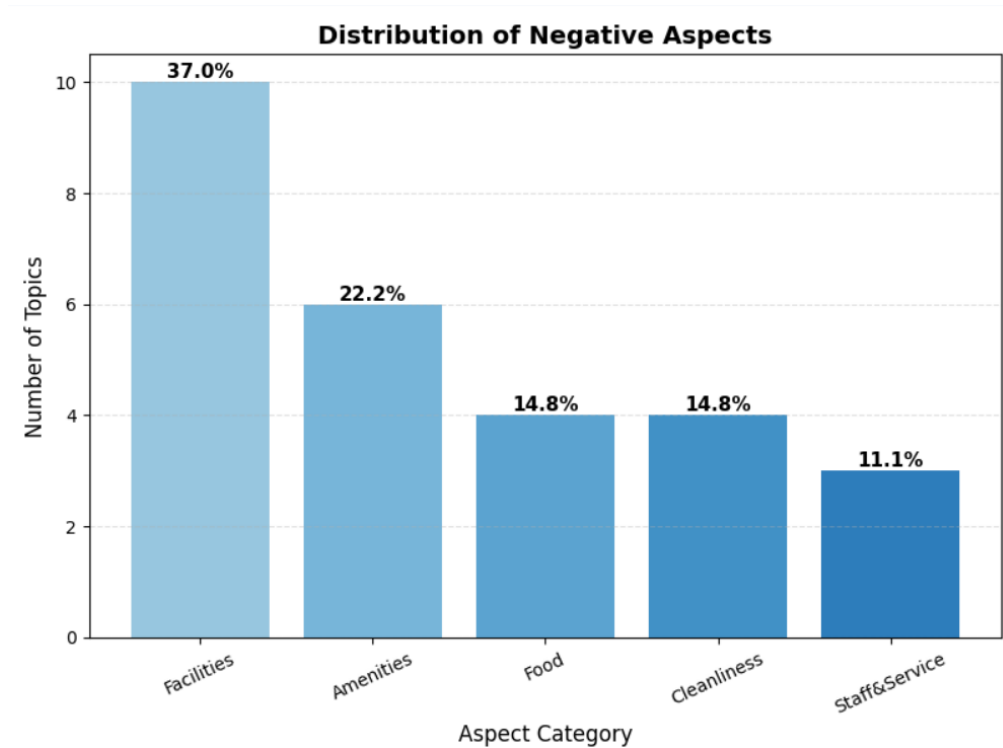


Figure 5.2: Frequency distribution of tourists' negative reviews by aspect

The aspect group requiring prioritized improvement and development is the Facilities group, with a frequency of occurrence in tourists' negative reviews at 37.0%, concentrating the most negative feedback related to rooms, furniture, and equipment. The dense appearance of keywords such as “chật” (cramped), “cũ” (old), “hỏng”

(broken), “điện” (electricity), “giường” (bed), or “cửa” (door)... is not merely a listing of defective items, but reflects a profound reality of asset value depreciation. The root cause does not lie in the negligence of a single employee, but originates from the asset life-cycle management strategy of the leadership or business owners. The application of a “reactive” maintenance mindset-only repairing upon breakdown instead of periodic maintenance and life-cycle upgrading, has left the quality of the accommodation space in a patchwork state. Consequently, the most fundamental and core needs of tourists, such as having a good night’s sleep on a comfortable bed or a convenient living space, are not met as expected. This directly erodes the luxury brand positioning that the hotel is striving to build.

Following the issues related to the accommodation space, the Amenities group accounts for 22.2%, also revealing fractures in the effort to ensure an uninterrupted experience for tourists. The convergence of negative feedback revolving around “Wifi yếu” (weak Wi-Fi), “hồ bơi nhỏ” (small swimming pool), or “check-in chậm” (slow check-in procedures) reflects direct interruptions in the service utilization process. These factors significantly reduce the comfort level and perceived value of customers, especially in the context of high-end accommodation services, where continuous and uninterrupted experiences play a core role. These manifestations expose a fundamental loophole in the responsiveness and forecasting capacity of the service system. The scale of utility infrastructure and operational procedures are only suitable for average capacity, lacking the capability to handle peak periods. When demand surges, the system quickly falls into overload, leading to a simultaneous decline in the overall service quality of the entire hotel system. These results reflect structural operational issues, specifically the absence of flexible coordination mechanisms and contingency plans. Failure to design a system capable of adapting to fluctuations will prevent the hotel from maintaining stability and consistency in its service quality.

Besides facilities and amenities, the Cleanliness aspect accounts for up to 14.8%, although it does not represent a high proportion of the feedback volume, it reveals “fatal” risks to the positioning of a luxury hotel. The appearance of sensitive keywords such as “mùi” (odor), “ấm mốc” (mold), “bụi” (dust), or “bẩn” (dirty) indicates inconsistencies in the cleaning process between different rooms or timeframes, exposing a severe lack of synchronization in the hotel’s space cleaning procedures. The issue points to

weaknesses in the execution of standard operating procedures (Housekeeping SOPs) and ineffectiveness in the quality control mechanism (Quality Assurance - QA) (Hayes, D.K., & Ninemeier, J.D., 2009). The existence of basic hygiene errors implies that the inspection and acceptance process has not been strictly implemented. Superficial or shallow supervisory activities lead to errors not being detected and addressed promptly. In the context of the hospitality industry, cleanliness is considered a foundational standard with a near-zero error tolerance level. Experiences involving foul odors or mold not only cause physical discomfort but also directly diminish customers' sense of safety and trust in the overall service quality.

Having an equivalent level of impact as the cleanliness group, the Food aspect is not a widespread bottleneck but directly affects brand valuation in the customers' minds. Keywords such as "buffet", "bữa sáng", "ít món" (few dishes), and especially the phrase "không xứng đáng với số tiền" (not worth the money) reflect a clear imbalance in "Perceived Value" between the costs they pay and the actual utilities received. These reviews do not merely criticize the taste of the food but weigh the expenses incurred for a premium standard against the actual quality received. Positioned in the high-end segment, breakfast is not just about providing nutrition but is part of the local cultural experience and luxury chain. The failure to create a culinary experience commensurate with the price point may be the consequence of excessive cost optimization or weaknesses in menu design techniques. The inability of the F&B (Food & Beverage) system to demonstrate diversity and refinement will leave customers feeling that their financial investment for the vacation has been significantly diminished.

Distinct from structural loopholes, the Staff & Service aspect appears at a lower frequency of approximately 11.1% and disperses into isolated situational incidents rather than forming a systemic problem cluster. Feedback related to procedural delays or inconsistent service attitudes indicates the presence of service variability, an unavoidable characteristic when quality depends on human factors. Although the data indicates that personnel quality overall has not fallen into a widespread decline, instability at critical moments such as check-in procedures or complaint handling still poses significant risks to the overall experience. These are touchpoints that directly influence customer evaluations, so any deviation in service quality can degrade satisfaction levels. These manifestations suggest that basic procedural systems have

been established, but the capacity to control interaction quality and standardize the service behaviors of frontline staff remains limited. The staff has not maintained consistency under high-pressure conditions or when out-of-process situations arise, thereby reducing the stability of service quality.

Overall, the aspects requiring improvement not only reflect isolated issues but demonstrate the existence of systemic limitations in asset management, service operations, and quality control. These are the primary factors degrading the consistency of the customer experience.

5.2.2. Comparison with 5-Star Hotel Standards

Based on the service quality limitations identified from empirical tourist review data, this study conducts a comparative analysis against the criteria outlined in the National Standard TCVN 4391:2015. This aims to elucidate the disparity between the current service quality of 4-star hotels and the requirements of the 5-star segment, thereby proposing effective improvement strategies. This comparison does not aim to evaluate the “pass or fail” compliance with the standards, but rather focuses on identifying systemic gaps-the factors currently hindering the upgrading potential of 4-star hotels in Vung Tau.

First, the gap in facilities is clearly manifested through feedback regarding “phòng chật” (cramped rooms), “cách âm kém” (poor soundproofing) or “cũ” and “hỏng” equipment. While the 5-star standard mandates spacious accommodation, high-quality materials and furnishings, as well as stable and synchronized operations, reality shows that many 4-star hotels have not maintained the optimal technical condition of their assets. This gap stems not only from the initial investment level but primarily lies in the capacity for asset life-cycle management and maintenance systems, leading to quality degradation over time.

Second, the gap in amenities is reflected through issues such as “wifi yếu” “hồ bơi quá tải” or an inefficient check-in process. The 5-star standard sets high requirements for the readiness and stability of the entire utility system at all times. Conversely, based on empirical data, the service systems at 4-star hotels are not yet designed to accommodate demand fluctuations, particularly during peak periods. This

indicates limitations in capacity design and resource coordination capabilities, which disrupts the customer experience.

Third, the gap in Cleanliness represents one of the most pronounced standard deviations. Feedback regarding “mùi hôi”, “ấm mốc”, “bụi bẩn” (foul odors, mold, and dirt) indicates that output quality control has not achieved a stable level. While 5-star standards demand a high degree of sanitation and tolerate zero errors, reality demonstrates that operational procedures at 4-star hotels still exhibit loopholes in the inspection and supervision phases. The core issue does not lie in the basic procedures, but rather in execution capacity and consistent quality assurance mechanisms.

Fourth, the gap in Food & Beverage is manifested through the disproportion between price and perceived value. According to the regulations of TCVN 4391:2015, 5-star catering services are highly stringent, requiring “a minimum of 2 restaurants serving diverse European and Asian cuisines; high-quality food and premium beverages.” Feedback describing the buffet as having ít món” or “không đáng tiền” (few dishes or being not worth the money) indicates that the dining service has not met the requirements for diversity and quality corresponding to the 5-star standard. This reflects limitations in menu design and value positioning, as the Food & Beverage service has not yet functioned as an experience-enhancing component, but rather remains supplementary and fails to generate a “perceived value” commensurate with the amount customers are willing to pay for a luxury vacation.

Finally, the gap in Staff & Service does not manifest as a systemic crisis but is evident in the lack of consistency during specific situations. Regarding human resources, the national standard requires 5-star hotel staff to possess a professional service style, a consistently friendly and attentive attitude, and flexible problem-solving abilities. Feedback regarding poor attitude, delays, or inflexible handling indicates that the degree of standardization in service behavior remains low. At the 5-star level, employees must not only strictly adhere to SOPs but also possess a service mindset to empathize with and personalize the experience, transforming rigid regulations into refined service; however, reality shows that the staff at 4-star hotels have not yet maintained uniform quality across all operational conditions.

The comparison with TCVN 4391:2015 demonstrates that the bottlenecks detected by the BERTopic model are not merely the subjective perceptions of a demanding customer segment, but essentially objective warnings regarding the lag behind the legal and professional standard frameworks. The identified gaps reflect not only disparities in investment levels but primarily reside in the capabilities of systemic administration, asset management, service design, and quality control. These are the core barriers preventing 4-star hotels in Vung Tau from approaching or transitioning to the 5-star segment, even if certain individual elements have achieved a relatively sufficient level.

5.3. Proposed Solutions for Service Quality Improvement

Based on the analysis of service aspects through Word Cloud visualization, key issues affecting customer experience have been identified. Accordingly, several practical solutions are proposed to address these limitations and improve overall service quality.

For facilities, hotels should prioritize upgrading and renovating deteriorated rooms, optimizing spatial layout to enhance comfort, and ensuring the provision of essential amenities. Pricing strategies should also be aligned with service quality and location. For amenities, efforts should focus on improving Wi-Fi performance, maintaining facilities regularly, and streamlining check-in and check-out procedures to reduce waiting time. Regarding food services, hotels should diversify menus, particularly breakfast buffets, improve food quality, ensure food safety standards, and adjust pricing to better reflect service quality. For staff and service, training programs should be implemented to enhance communication skills and professionalism. Workforce allocation during peak hours should be optimized, and service processes should be standardized. Additionally, a responsive feedback system should be established to handle customer requests and complaints effectively.

Finally, for cleanliness, strict housekeeping procedures should be enforced through detailed checklists, issues related to odors and mold should be resolved, cleaning frequency should be increased, and hygiene-sensitive items should be replaced regularly to ensure a safe and clean environment.

5.4. Limitations and Future Research Directions

5.4.1. Limitations

Although the research has achieved certain results in constructing a sentiment classification model and extracting information from review data of 4-star hotels by tourists on the Traveloka platform, several limitations still exist and need to be considered to enhance the reliability and generalizability of the findings.

First, natural language data preprocessing still has certain limitations. Text normalization and cleaning have not thoroughly addressed multilingual cases, particularly, Chinese reviews have not been effectively translated and normalized. Furthermore, the phenomenon of code-mixing between Vietnamese and English within the same sentence (e.g., “phòng clean, view đẹp”, “service ok nhưng hơi slow”) has not been completely resolved, affecting the quality of text representation. Elements such as emojis and teen code have also not been thoroughly cleaned due to the reliance on manual processing methods; consequently, some common slang terms like “okieeeeeeeee”, “okie”, “xịn xò”, etc., can still introduce noise into the model.

Besides, during the data labeling process, the study adopted a binary sentiment classification approach (positive and negative) to mitigate the issue of data imbalance among classes. Although this solution helps optimize the predictive performance of machine learning algorithms, the omission of the “Neutral” label inadvertently eliminates a crucial dimensional space of information. This trade-off causes the model to miss objective feedback, constructive reviews, or especially reviews with “mixed sentiments”, which praise one aspect while complaining about another.

Another notable limitation relates to the service aspect-based analysis phase. Although the research categorized reviews into specific aspects (such as Facilities, Amenities, Cleanliness, Food, Staff & Service), the obtained results remain somewhat ambiguous and contain a high level of noise. Specifically, within certain aspects such as “Food”, the representative keyword set encompasses not only words related to food or buffet but also includes terms belonging to other aspects such as television or beach. This indicates that the model has not distinctly separated the topics, leading to aspect overlapping and reducing the interpretability of the results from a managerial perspective.

Regarding the scope and data sources, the study solely utilized data from a single platform, Traveloka, and focused on the 4-star hotel segment in Vung Tau. This limits the representativeness and generalizability of the research sample, as user behavior and reviewing trends may differ across platforms (such as Booking, Agoda, or Google Reviews). Simultaneously, the uneven distribution of review volumes among hotels on the same platform can create a certain bias, leading to potential deviations in evaluating the overall picture of service quality across the entire market.

5.4.2. Future Research Directions

Based on the analyzed limitations, this study proposes several directions for future development and expansion to refine the model and enhance its practical value. First, the data collection process needs to be improved by expanding data sources across various platforms such as Agoda, Booking.com, etc., to ensure representativeness and mitigate platform-specific biases. Future research could be broadened by integrating multi-source data and conducting comparative analyses across platforms, thereby evaluating disparities in customer behavior and reviewing trends.

Furthermore, it is essential to enhance the quality of natural language data preprocessing, particularly in multilingual processing and the normalization of non-standard texts. In subsequent studies, high-precision machine translation models such as neural machine translation (NMT) could be integrated, or multilingual language models like mBERT or XLM-R could be utilized directly to process data without requiring complete translation into Vietnamese. This approach mitigates semantic deviation during the translation process, particularly for sentences containing highly tourism-specific contexts. Another advanced approach involves directly integrating preprocessing steps into the deep learning pipeline, rather than relying entirely on manual processing. This can be combined with data augmentation techniques such as back-translation or synonym replacement to make the model more robust against non-standard linguistic variations.

To overcome the information deficit from neutral reviews and optimize the value of input data, the label space should be re-evaluated by reverting to a multi-class classification problem with all three labels: positive, negative, and neutral. Rather than eliminating the minority class to circumvent data imbalance, future research could adopt

more advanced data balancing techniques in NLP. Specifically, textual data augmentation methods, SMOTE, or direct fine-tuning of the model's loss function could be employed to compel the algorithm to focus on learning the features of the neutral and constructive feedback datasets.

To comprehensively resolve the challenge posed by reviews with “mixed sentiments”, a breakthrough developmental direction is to transition from document-level sentiment analysis to Aspect-Based Sentiment Analysis. This technique enables the model to granulate and extract multiple service entities mentioned within a single complex sentence, assigning independent sentiment labels to each respective entity. This paradigm shift not only preserves the complete informational dimensionality but also equips hotel managers with a micro-measurement toolkit, accurately pinpointing the “bottlenecks” requiring improvement within the service value chain.

REFERENCES

- Jurafsky, D., & Martin, J. H. (2024). Speech and language processing (3rd ed., draft).
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2023). An introduction to statistical learning: With applications in Python (2nd ed.).
- Raschka, S., Liu, Y., Mirjalili, V., & Dzhulgakov, D. (2022). Machine learning with PyTorch and scikit-learn: Develop machine learning and deep learning models with scikit-learn and PyTorch (2nd ed.).
- Trương Quang Nhật. (n.d.). Sử dụng phương pháp Support Vector Machines (SVMs) trong mô hình học máy. Khoa Toán Kinh tế, Trường Đại học Kinh tế – Luật.
- Pham Dinh Khanh.(2020).Đánh giá mô hình phân loại ML.Khoa học dữ liệu - Khanh's blog. <https://phamdinhkhanh.github.io/2020/08/13/ModelMetric.html>
- Kohavi, R. (1995). A study of cross-validation and bootstrap for accuracy estimation and model selection. *Proceedings of the 14th International Joint Conference on Artificial Intelligence (IJCAI)*, 2, 1137-1143.
https://www.researchgate.net/publication/2352264_A_Study_of_Cross-Validation_and_Bootstrap_for_Accuracy_Estimation_and_Model_Selection
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.
https://www.researchgate.net/publication/51969319_Scikit-learn_Machine_Learning_in_Python
- Cronin, J. J., & Taylor, S. A. (1992). Measuring Service Quality: A Reexamination and Extension. *Journal of Marketing*, 56(3), 55-68.
https://www.researchgate.net/publication/225083621_Measuring_Service_Quality_-_A_Reexamination_And_Extension

Bộ Khoa học và Công nghệ. (2015). *TCVN 4391:2015 Khách sạn - Xếp hạng* (Hotel - Classification). Hà Nội.

[https://vhthdl.daklak.gov.vn/Upload/Images/2-TCVN-4391-2015\(1\).pdf](https://vhthdl.daklak.gov.vn/Upload/Images/2-TCVN-4391-2015(1).pdf)

Hayes, D. K., & Ninemeier, J. D. (2009). *Hotel Operations Management*.

Potdar, K., Pardawala, T. S., & Pai, C. D. (2017). A Comparative Study of Categorical Variable Encoding Techniques for Neural Network Classifiers. *International Journal of Computer Applications*, 175(4), 7-15.

Jurafsky, D., & Martin, J. H. (2009). *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition* (2nd ed.). Pearson Prentice Hall.

Dahouda, M. K., & Joe, I. (2021). A Deep-Learned Embedding Technique for Categorical Features Encoding. *IEEE Access*, 9, 114012-114024. doi:10.1109/ACCESS.2021.3104357

Wang, Y., Li, J., Yang, B., Song, D., & Zhou, L. (2024). Orthogonal Matrix-Autoencoder-Based Encoding Method for Unordered Multi-Categorical Variables with Application to Neural Network Target Prediction Problems. *Applied Sciences*, 14, 7466. <https://doi.org/10.3390/app14177466>

Bafna, P., Pramod, D., & Vaidya, A. (2016). Document Clustering: TF-IDF approach. Trong Hội nghị Quốc tế về Kỹ thuật Điện, Điện tử và Tối ưu hóa (ICEEOT).

Zhang, W., Yoshida, T., & Tang, X. (2011). A comparative study of TFIDF, LSI and multi-words for text classification. *Expert Systems with Applications*, 38, 2758-2765.

Cronin, J. J., & Taylor, S. A. (1992). Measuring Service Quality: A Reexamination and Extension. *Journal of Marketing*, 56(3), 55-68.

https://www.researchgate.net/publication/225083621_Measuring_Service_Quality_-_A_Reexamination_And_Extension

Bộ Khoa học và Công nghệ. (2015). *TCVN 4391:2015 Khách sạn - Xếp hạng* (Hotel - Classification). Hà Nội.

[https://vhthdl.daklak.gov.vn/Upload/Images/2-TCVN-4391-2015\(1\).pdf](https://vhthdl.daklak.gov.vn/Upload/Images/2-TCVN-4391-2015(1).pdf)

Nguyễn, V. B., & Nguyễn, T. H. Y. (2023). *Tích hợp lý thuyết U&G và lý thuyết hành vi hoạch định để đánh giá tác động của quảng cáo Facebook đối với ý định mua hàng*. Tạp chí Thương mại Khoa học, (173), 65–80.

<https://www.brandsvietnam.com/congdong/topic/334589-mo-hinh-elm-elaboration-likelihood-model-trong-quang-cao>

McInnes, L., Healy, J., & Melville, J. (2018). *UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction*. arXiv:1802.03426.

Reimers, N., & Gurevych, I. (2019). *Sentence-BERT: Sentence embeddings using Siamese BERT-networks*.